Anomaly Detection in Satellite-Borne Push-Broom Hyperspectral Imagery Based on Joint Low-Rank Tensor Approximation

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***Abstract*— Hyperspectral anomaly detection aims to detect potential targets of interest in hyperspectral images (HSIs). In this article, we propose a joint low-rank tensor approximation-based anomaly detection method that incorporates the distinct physical characteristics of satellite-borne push-broom HSIs. In our method, an HSI is decomposed into a background tensor and an anomaly tensor. For the background tensor, the spatial correlations between pixels give it a low-rank characteristic in frontal slices, while the spectral correlations among adjacent bands and similar ground materials give it low-rank properties in horizontal slices. Therefore, in order to simultaneously and flexibly exploit the correlations in both the spatial and spectral dimensions of the background tensor, we design a joint low-rank constraint term of the background tensor, using the weighted tensor nuclear norm based on tensor singular value decomposition (t-SVD), which applies to both the background tensor and its permutation tensor. In addition, considering the homogeneity of adjacent pixels in the background, which results in spatial piecewise smoothness of the background tensor, we incorporate a linear total variation norm regularization term to more accurately characterize the background tensor. For the anomaly tensor, an $l_{1,1,2}$-norm regularization term is employed to characterize its tubewise sparsity. We incorporate all regularization terms and a fidelity term into a nonconvex framework and solve it using the alternating direction method of multipliers. The experimental results using real HSIs from the GaoFen-5 and ZiYuan1-02D satellites demonstrated that the proposed method outperforms seven comparison methods.**

***Index Terms*— Hyperspectral anomaly detection, push-broom imaging, tensor decomposition, weighted tensor nuclear norm.**

NOMENCLATURE

$x, \boldsymbol{x}, \boldsymbol{X}, \boldsymbol{\mathcal{X}}$	Scalar, vector, matrix, and tensor.
$\boldsymbol{\mathcal{X}}(i, :, :)$	ith horizontal slice.
$\boldsymbol{\mathcal{X}}(:, i, :)$	ith lateral slice.
$\boldsymbol{\mathcal{X}}(:, :, i)$	ith frontal slice.
$\boldsymbol{\mathcal{X}}(i, j, :)$	(i, j)th tube fiber.
$\boldsymbol{X}^{(i)}$	$\boldsymbol{\mathcal{X}}(:, :, i)$.
$\boldsymbol{\mathcal{X}}^T$	Transpose of $\boldsymbol{\mathcal{X}}$.
$\boldsymbol{\mathcal{X}}_p$	permute($\boldsymbol{\mathcal{X}}$, [3, 2, 1]).
$\boldsymbol{X}_{(n)}$	Mode-n matricization of $\boldsymbol{\mathcal{X}}$.
$\overline{\boldsymbol{\mathcal{X}}}$	fft($\boldsymbol{\mathcal{X}}$, [], 3).
$\|\boldsymbol{\mathcal{X}}\|_F$	Tensor Frobenius norm, $\Sigma_{i,j}\|\boldsymbol{\mathcal{X}}(i, j, :)\|_F$.
$\| \boldsymbol{\mathcal{X}} \|_{\circledast}$	Tensor nuclear norm.
$\| \boldsymbol{\mathcal{X}} \|_{\boldsymbol{w},\circledast}$	Weighted tensor nuclear norm.

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I. INTRODUCTION

HYPERSPECTRAL images (HSIs) not only encompass the spatial information of various materials but also contain rich spectral information, referred to as data cubes. The spectral characteristics of the hundreds of bands in HSIs enable them to distinguish subtle differences between objects, which is a challenging task when solely using spatial information. In the past few decades, HSIs have been widely utilized in various applications, such as classification [1], [2], mineral surveys [3], [4], gas emission detection [5], [6], target detection [7], [8], and anomaly detection [9], [10].

Anomaly detection aims to highlight anomalies and suppress the background without utilizing prior spectral knowledge of the anomaly targets. It serves as an important task for providing potential regions of interest for subsequent target detection and recognition. Generally, hyperspectral anomaly detection methods rely on two fundamental characteristics of the anomalous targets in HSIs: 1) anomalies exhibit differences from their neighborhood background in terms of both spectral and spatial features and 2) anomalies occur in HSIs with low probability, occupying fewer pixels.

In recent decades, many effective hyperspectral anomaly detection methods have been proposed. The statistic-based Reed–Xiaoli (RX) algorithm [11] is a widely used classical anomaly detection method that is considered as a benchmark. The RX algorithm is based on the fundamental assumption that the background pixels follow a multivariate Gaussian distribution. After estimating the mean vector and covariance matrix of the background pixels, the Mahalanobis distance between the pixel under test and the mean vector is used to obtain the anomaly values. In subsequent research, two variants of the RX algorithm were developed: 1) the global RX algorithm [12] that estimates the mean and covariance matrix using all pixels in the HSI and 2) the local RX algorithm [13] that estimates statistics within a sliding window. However, two limitations

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restrict the detection performance of the RX algorithm for real HSIs. First, real HSIs often do not follow simple multivariate Gaussian distribution. Second, anomalies and noise in HSIs can interfere with the estimation of statistics. Therefore, many improved algorithms based on the RX algorithm have been proposed. Kown and Nasrabadi [14] proposed the KRX algorithm that employs a kernel function to map the original hyperspectral data into a high-dimensional feature space, thus enabling the modeling of more complex background distributions. To avoid contamination by anomalous and noisy pixels in the background statistics estimation, Du and Zhang [15] proposed a random selection-based anomaly detector (RSAD). This method involves repeatedly performing a random selection of background samples to complete the estimation of the background statistics. Guo et al. [16] proposed a weighted RX detector that reduces the weights of potentially anomalous pixels and increases the weights of the potential background pixels. However, it is difficult to establish a unified statistical model for many real HSIs, limiting the application of such methods.

Representation-based methods have been widely studied and applied to hyperspectral anomaly detection. The sparse representation (SR) model [17] is fundamental to such methods. The basic concept is that background pixels can be represented using atoms from an overcomplete dictionary with a small residual representation, whereas anomalous pixels have larger residual representations. The representation residual of each pixel under testing can be used as the anomaly value. Many improved methods based on SR have been proposed. Li et al. [18] used a dual-window strategy to obtain a purer dictionary and combined it with a joint SR model, proposing a background joint SR detector that improves the detection performance of SR methods. To ensure the physical meaning of the model, Ling et al. [19] imposed nonnegative and sum-to-one constraints on the coefficient vector in the proposed SR and linear mixture model (SR-LMM)-based detector, while removing the upper bound constraint on sparsity. The sparsity score estimation anomaly detection framework calculates the frequency of each dictionary atom and transforms the frequencies into sparsity scores to obtain the anomaly values. Another classic representation-based device for anomaly detection is a collaborative representation-based detector (CRD) [20]. A CRD represents a pixel under test using its neighboring pixels and utilizes the representation residual as the anomaly value. However, the core of such representation-based methods is linear modeling, and the nonlinear characteristics in real HSIs limit the performance of these methods.

Deep-learning-based methods for hyperspectral anomaly detection have also attracted significant attention. For supervised methods, Li et al. [21] proposed a transferred deep convolutional neural network to generate the difference between pixel pairs. During network training, the labels for similar pixel pairs are set to zero, whereas the labels for dissimilar pixel pairs are set to one. When performing detection, the pixel pairs consisting of pixels under testing and its surrounding pixels are fed into the trained network, which generates similarity scores. The average of these similarity scores is used as the anomaly value for the pixel under testing. The practical application of supervised deep-learning-based methods is limited by the availability of labeled samples. Therefore, unsupervised deep-learning-based methods may be more suitable for hyperspectral anomaly detection. In general, unsupervised deep-learning-based methods, such as autoencoder network-based methods [22], [23] and generative adversarial network-based methods [24], [25], reconstruct pixels using the network and utilize the residuals between the reconstructed and original pixels as anomaly values. However, both supervised and unsupervised deep learning methods face the issue of lacking sufficiently large training HSI datasets.

Decomposition-based methods have been widely applied to detect hyperspectral anomalies, recently. The basic idea of these methods is to decompose the original HSI into the background and anomalous components based on their intrinsic characteristics. Depending on whether the HSI cube unfolds into a matrix during the decomposition process, hyperspectral anomaly detection methods can be divided into two categories: matrix decomposition- and tensor decomposition-based anomaly detection methods. For matrix decomposition-based methods, low-rank and sparse matrix decomposition (LRaSMD) is a classic and commonly used technique that assumes that the background matrix is low-rank and the anomaly matrix is sparse. Sun et al. [26], within the anomaly matrix obtained using LRaSMD, calculated the Euclidean distance between the pixel under test and the mean vector to obtain the anomaly value. Inspired by LRaSMD, Zhang et al. [27] proposed an LRaSMD-based Mahalanobis distance (LSMAD) method, in which they calculate the Mahalanobis distance on the decomposed background matrix to obtain an anomaly detection map. To obtain a more accurate background matrix, the low-rank and SR (LRASR) method proposed in [28] employs the K-means algorithm to build a dictionary and represents the background matrix as the product of the dictionary and the coefficient matrix. In addition, a sparse regularization term for the coefficient matrix is added to the optimization model. To preserve the geometrical structure and spatial relationships, Cheng and Wang [29] proposed a graph and total variation-regularized LRR (GTVLRR) method. This method incorporates variation regularization and graph regularization terms into the low-rank regression model.

Because an HSI is a data cube, matrix decomposition-based methods unfold it into a matrix that disrupts its intrinsic 3-D structure. In comparison, tensor-decomposition-based methods represent HSIs using third-order tensors, preserving the inherent 3-D structure. Li et al. [30] proposed a prior-based tensor approximation (PTA) method that applies low-rank and smoothness constraints to the background tensor and a sparse constraint to the anomaly tensor. However, in the PTA method, the low-rank constraint on the background tensor is actually imposed on the mode-3 unfolding matrix of the background tensor. This approach is similar to matrix decomposition methods and may not fully explore the inherent 3-D structure of HSIs. Directly imposing a low-rank constraint on the background tensor preserves this inherent 3-D structure of HSIs. The most crucial question in the tensor low-rank decomposition model is how to characterize the rank of a

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tensor. Unlike the definition of the matrix rank, that of the tensor rank is not unique. Typically, the definition of the tensor rank depends on the specific tensor decomposition form used [31].

The conventional tensor decomposition models, including Tucker decomposition [32], CANDECOMP/PARAFAC (CP) decomposition [33], and tensor singular value decomposition (t-SVD) [34], have been widely applied. Zhao et al. [35] created a third-order tensor through multiple morphological profiles and employed Tucker decomposition to estimate and eliminate the background, thereby achieving satisfactory small target detection. Removing the major components through Tucker decomposition to retain small targets is essentially similar to imposing low-rank constraints on the background tensor to remove the background tensor and obtain the target. Xue et al. [36] applied the CP decomposition to low-rank tensor completion and proposed a multilayer sparse regularization term to adequately describe the structured sparsity of tensors. Many hyperspectral anomaly detection methods based on these tensor decomposition models have achieved excellent performance. Li et al. [37] applied the Tucker decomposition to hyperspectral anomaly detection and proposed a low-rank tensor decomposition-based anomaly detection (LTDD) algorithm. In [38], an abundance tensor regularization with low rankness and smoothness based on sparse unmixing (ATLSS) is proposed for hyperspectral anomaly detection. The ATLSS method characterizes the low rankness of the background abundance tensor by utilizing CP decomposition. However, CP decomposition struggles to capture the correlations across different dimensions of the tensor flexibly, and the determination of the CP rank is a nondeterministic polynomial. Tucker decomposition disrupts the internal structure of a tensor by unfolding it into a matrix along different dimensions. These limitations hinder the performance of the corresponding anomaly detection methods.

The t-SVD is a novel tensor decomposition method, which has a structure similar to matrix SVD, and it replaces matrix multiplication between matrices with the t-product [34] between tensors. It can characterize correlations within frontal slices using SVD and describe correlations in the third dimension through circular convolution between tube fibers. Therefore, utilizing the tensor multirank [39] defined by t-SVD allows the characterization of low-rank properties in frontal slices of the tensor while also incorporating the correlations along the tube fibers. Many hyperspectral anomaly detection methods based on tensor low-rank decomposition using t-SVD have been proposed, such as in [40] and [41].

However, the aforementioned anomaly detection methods primarily focus on the methods themselves. Applying them to anomaly detection in HSIs captured through push-broom imaging may not achieve the desired performance. Push-broom imaging is a commonly employed hyperspectral imaging technique. The push-broom imaging method results in an HSI with certain inherent spatial–spectral characteristics.

In this study, considering the inherent spatial–spectral characteristics of HSIs obtained through satellite push-broom imaging, we propose an anomaly detection method based on joint low-rank tensor approximation (JLRTA). In our method, the HSI is decomposed into background and anomaly tensors. In order to simultaneously and flexibly characterize the spatial dimensional low rankness in the frontal slices and the spectral dimensional low rankness in the horizontal slices introduced by push-broom imaging, we impose a joint low-rank constraint on the background tensor. Specifically, the weighted tensor nuclear norm based on t-SVD is used as a measure of the tensor rank. We use the weighted tensor nuclear norm of the original HSI tensor as the low-rank constraint term in the spatial dimension and the weighted tensor nuclear norm of the permutation tensor of the original tensor as the low-rank constraint in the spectral dimension. The weighted sum of both is used as a joint low-rank constraint term. It can flexibly characterize the low-rank properties of both dimensions simultaneously. Furthermore, to characterize the spatial piecewise smoothness, a linear total variation (TV) norm regularization term is incorporated. For the anomaly tensor, an $l_{1,1,2}$-norm regularization term is utilized to promote tubewise sparsity. By applying the alternating direction method of multipliers (ADMM), the decomposition problem is solved to obtain the optimal background and anomaly tensors. Finally, the anomaly map is obtained from the anomaly tensor. An overall flowchart of the proposed JLRTA anomaly detection method for satellite-borne push-broom HSI is shown in Fig. 1.

The main contributions of this study can be summarized as follows.

1) Considering the inherent spatial–spectral characteristics of push-broom HSIs, a joint low-rank regularization term for the background tensor is designed and incorporated into the decomposition model. This regularization term simultaneously and flexibly constrains the low rankness in frontal slices caused by spatial correlations and the low rankness in horizontal slices caused by spectral correlations. A spatial piecewise smoothness regularization term for the background tensor and a tubewise sparsity regularization term for the anomaly tensor are also incorporated into the model. This helps to better separate the anomalies from the background tensor.
2) We use the weighted tensor nuclear norm based on t-SVD as the measure of the tensor rank in a certain dimension and combine it with tensor permutation operations to obtain the joint low-rank regularization term. In addition, we employ the linear TV-norm as the spatial piecewise smoothness regularization term and use the $\ell_{1,1,2}$-norm the tubewise sparsity regularization term.
3) Based on the ADMM framework, an algorithm was designed to solve the proposed model. We conducted experiments on real push-broom HSIs acquired from the GaoFen-5 (GF-5) and ZiYuan1-02D (ZY1-O2D) satellites. The experimental results demonstrate that the proposed method outperforms several comparative methods across various evaluation metrics.

The remainder of this article is organized as follows. Section II introduces the notations and preliminaries used in this study. Section III describes real satellite-borne push-broom HSI datasets, and Section IV provides a detailed description of the proposed method and the corresponding optimization

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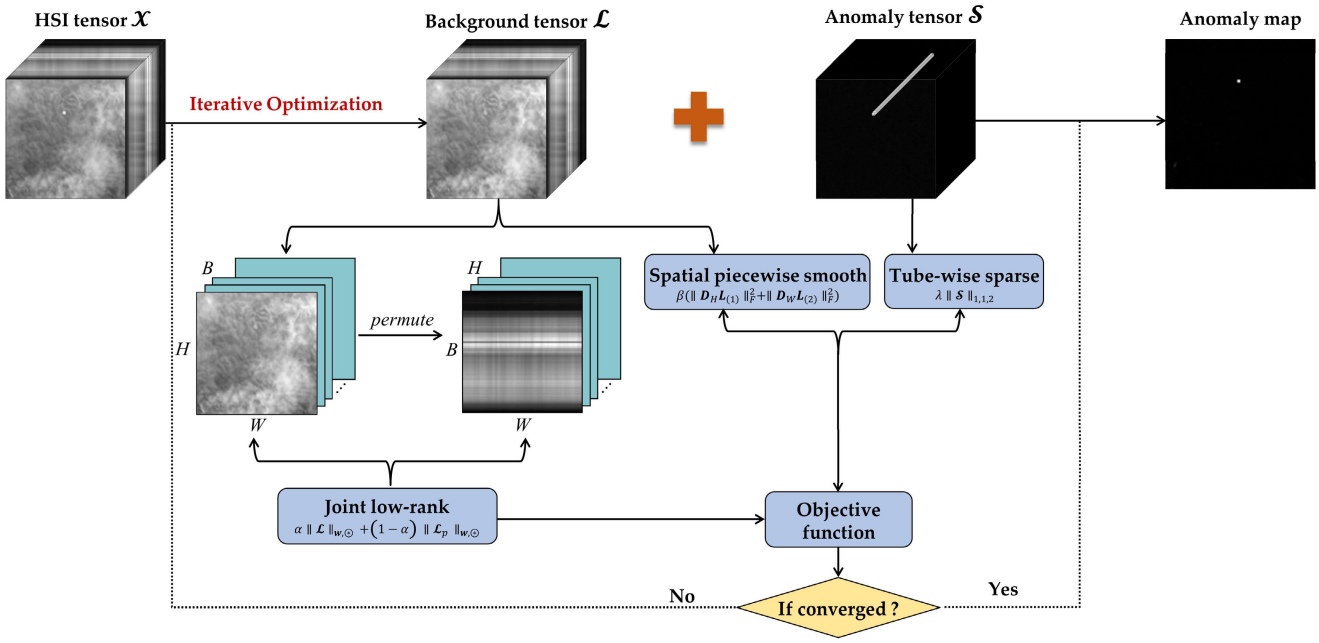


Fig. 1. Flowchart of the proposed JLRTA hyperspectral anomaly detection method.

algorithm. The experimental results and discussion analysis are presented in Section V. Finally, the conclusions are presented in Section VI.

II. Notations and Preliminaries

A. Notations

In this article, tensors are denoted by bold Euler script letters, for example, $\mathcal{X}$. Matrices are represented by bold capital letters, for example, $\boldsymbol{X}$. Vectors are indicated by bold lowercase letters, for example, $\boldsymbol{x}$. Scalars are denoted by lowercase letters, for example, x. Slices are 2-D sections of a tensor determined by fixing all but two indices. Fibers are higher order analogs of matrix rows and columns defined by fixing all but one index [31]. For a 3-D tensor $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ representing an HSI, the notations $\mathcal{X}(:,:,i)$, $\mathcal{X}(:,i,:)$, and $\mathcal{X}(i,:,:)$ represent the ith frontal, lateral, and horizontal slices, respectively. The notations $\mathcal{X}(:,i,j)$, $\mathcal{X}(i,:,j)$, and $\mathcal{X}(i,j,:)$ indicate the (i,j)th mode-1, mode-2, and mode-3 fibers, respectively, where mode-3 fiber is also referred to tube fiber. We denote the permutation of $\mathcal{X}$ as $\mathcal{X}_p$, which is obtained by applying the permutation operation with order $[3,2,1]$ to $\mathcal{X}$, denoted as $\mathcal{X}_p = \text{permute}(\mathcal{X}, [3,2,1])$. This operation swaps the third and first dimensions of $\mathcal{X}$. Consequently, the frontal slice of $\mathcal{X}_p$ corresponds to a horizontal slice of $\mathcal{X}$. The mode-n matricization (unfolding) of tensor $\mathcal{X}$, denoted as $\boldsymbol{X}_{(n)}$, is the process of sequentially arranging mode-n fibers into the columns of a matrix [31]. This process can be expressed as $\boldsymbol{X}_{(n)} = \text{unfold}_n(\mathcal{X})$, and the reverse process is represented as $\mathcal{X} = \text{fold}_n(\boldsymbol{X}_{(n)})$. We use $\overline{\mathcal{X}}$ to represent the tensor obtained by performing the fast Fourier transform (FFT) on the mode-3 fibers of $\mathcal{X}$. This process can be expressed as $\overline{\mathcal{X}} = \text{fft}(\mathcal{X}, [], 3)$, and the inverse is represented as $\mathcal{X} = \text{ifft}(\overline{\mathcal{X}}, [], 3)$. For convenience, we summarize the frequently used notations in Nomenclature.

B. t-SVD

First, we introduce the t-product, which is a multiplication operation between third-order tensors that resembles matrix multiplication. A tensor of size $n_1 \times n_2 \times n_3$ can be regarded as a matrix of size $n_1 \times n_2$, where the matrix elements correspond to the tubes of the tensor.

For the t-product, the multiplication operation between the elements was replaced by a circular convolution between the tubes. The circular convolution operation is denoted by "$\star$." The t-product is defined as follows:

Definition 1 (t-Product [34]): The t-product $\mathcal{C} = \mathcal{A} * \mathcal{B}$ of $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ and $\mathcal{B} \in \mathbb{R}^{n_2 \times n_4 \times n_3}$ is a tensor of size $n_1 \times n_4 \times n_3$, where the (i,j)th tube denoted by $\mathcal{C}(i,j,:)$ for $i = 1, 2, \ldots, n_1$ and $j = 1, 2, \ldots, n_4$ of tensor $\mathcal{C}$ is given by $\sum_{k=1}^{n_2} \mathcal{A}(i,k,:) \star \mathcal{B}(k,j,:)$.

Next, we provide some related tensor notations.

Definition 2 (Tensor Transpose [34]): Let $\mathcal{A}$ be a tensor of size $n_1 \times n_2 \times n_3$. Then, $\mathcal{A}^T$ is the $n_2 \times n_1 \times n_3$ tensor obtained by transposing each of the frontal slices and reversing the order of the transposed frontal slices 2 through n_3.

Definition 3 (Identity Tensor [34]): The identity tensor $\mathcal{I} \in \mathbb{R}^{n \times n \times n_3}$ is the tensor whose first frontal slice is the $n \times n$ identity matrix and whose other frontal slices are all zeros.

Definition 4 (Orthogonal Tensor [34]): A tensor $\mathcal{Q} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ is orthogonal if

$$\mathcal{Q}^T * \mathcal{Q} = \mathcal{Q} * \mathcal{Q}^T = \mathcal{I} \tag{1}$$

where $*$ is the t-product.

Definition 5 (f-Diagonal Tensor [34]): A tensor is called f-diagonal if each frontal slice of the tensor is a diagonal matrix.

Then, we can define the t-SVD factorization based on the t-product.

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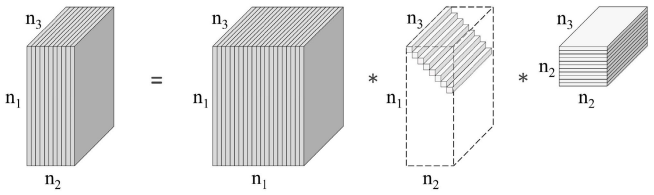


Fig. 2. Illustration of the t-SVD of an $n_1 \times n_2 \times n_3$ tensor.

Theorem 1 (t-SVD [34]): For $\boldsymbol{\mathcal{A}} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, it can be factored as

$$\boldsymbol{\mathcal{A}} = \boldsymbol{\mathcal{U}} * \boldsymbol{\mathcal{S}} * \boldsymbol{\mathcal{V}}^T \tag{2}$$

where $\boldsymbol{\mathcal{U}}$ and $\boldsymbol{\mathcal{V}}$ are orthogonal tensors with sizes $n_1 \times n_1 \times n_3$ and $n_2 \times n_2 \times n_3$, respectively, and $\boldsymbol{\mathcal{S}}$ is an $n_1 \times n_2 \times n_3$ f-diagonal tensor.

Fig. 2 illustrates t-SVD factorization. Due to the circular convolution operation of the t-product, t-SVD can be computed more efficiently using matrix SVD in the Fourier domain. For detailed information about the computation, please refer to [34] and [42]. Based on t-SVD factorization, Kilmer et al. [39] proposed a new measure of tensor rank complexity called tensor multirank.

Definition 6 (Tensor Multirank [39]): The multirank of $\boldsymbol{\mathcal{A}} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ is a vector $\boldsymbol{r} \in \mathbb{R}^{n_3 \times 1}$ with the ith element equal to the rank of ith frontal slice of $\overline{\boldsymbol{\mathcal{A}}}$ obtained by taking the Fourier transform along the third dimension of the tensor.

The multirank tensor is nonconvex. Therefore, for practical applications, Semerci et al. [43] proposed the tightest convex relaxation to ℓ_1-norm of the tensor multirank, which is called the tensor nuclear norm. In the following, we provide this definition.

Definition 7 (Tensor Nuclear Norm [43]): For $\boldsymbol{\mathcal{A}} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, the t-SVD-based tensor nuclear norm is defined as the sum of the singular values of all the frontal slices of $\overline{\boldsymbol{\mathcal{A}}}$ and is represented as

$$\| \boldsymbol{\mathcal{A}} \|_{\circledcirc} = \sum_{i=1}^{n_3} \| \overline{\boldsymbol{A}}^{(i)} \|_* = \sum_{i=1}^{n_3} \sum_{j=1}^{\min(n_1,n_2)} \sigma_j \left(\overline{\boldsymbol{A}}^{(i)} \right) \tag{3}$$

where $\sigma_j(\overline{\boldsymbol{A}}^{(i)})$ denotes the jth largest singular value of $\bar{A}^{(i)}$.

C. Satellite-Borne Push-Broom Hyperspectral Imaging

Push-broom imaging is a commonly used hyperspectral imaging technology. As shown in Fig. 3, the push-broom imaging technique utilizes an area-array detector to acquire images line-by-line. The spectral signal of each ground strip in the cross-track direction is dispersed using a spectrometer and imaged onto an area array detector [42]. The (2-D data acquired by the area-array detector in each frame correspond to a horizontal slice of the HSI tensor within the green square in Fig. 3. This slice represents the spectral information of a ground strip, with one dimension being the spatial dimension and the other being the spectral dimension. As the satellite moves along its orbit, imaging is performed strip-by-strip in the along-track direction. Multiple horizontal slices are stacked along the direction of the white arrow in Fig. 3 to generate a complete HSI, which is a data cube.

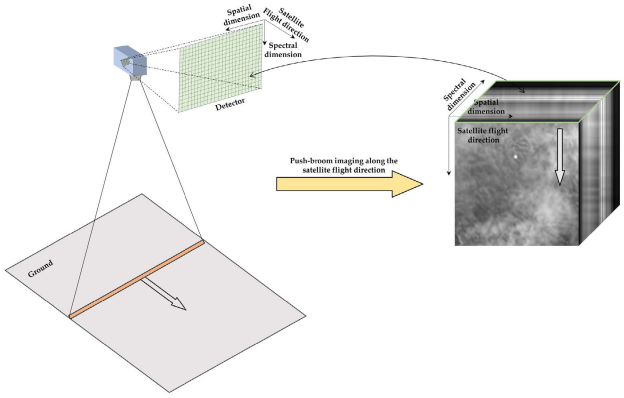


Fig. 3. Illustration of satellite-borne push-broom hyperspectral imaging.

TABLE I
MAIN INFORMATION ABOUT THE TWO AHSI SENSORS

Satellite/ Payload	Spectral range (nm)	Number of bands	Spectral resolution (nm)	Spatial resolution (m)	Swath width (km)
GF-5/ AHSI	400–1000 (VNIR) 1000–2500 (SWIR)	150 (VNIR) 180 (SWIR)	5 (VNIR) 10 (SWIR)	30	60
ZY1-02D/ AHSI	400–1000 (VNIR) 1000–2500 (SWIR)	76 (VNIR) 90 (SWIR)	10 (VNIR) 20 (SWIR)	30	60

III. HYPERSPECTRAL DATASETS

This section introduces two satellite-borne push-broom imaging hyperspectral datasets used for anomaly detection, along with the sources of the datasets: hyperspectral imagers aboard the GF-5 and ZiYuan-1-02D (ZY1-02D) satellites.

A. Advanced Hyperspectral Imager

The two hyperspectral datasets used in this study consisted of the HSIs captured by the Advanced Hyperspectral Imager (AHSI) aboard the GF-5 satellite [45] and the AHSI aboard the ZY1-02D satellite [46]. Both utilize push-broom imaging technology. The AHSI aboard the GF-5 satellite covers a spectral range of 400–2500 nm. It achieves a spectral resolution of 5 nm with 150 spectral bands in the visible and near-infrared (VNIR) regions and a spectral resolution of 10 nm with 180 spectral bands in the short-wave infrared (SWIR) region. It has a swath width of 60 km and a spatial resolution of 30 m. The AHSI aboard the ZY1-02D satellite also covers a spectral range of 400–2500 nm. It has a spectral resolution of 10 nm with 76 spectral bands in the VNIR region and a spectral resolution of 20 nm with 90 spectral bands in the SWIR region. It has a swath width of 60 km and a spatial resolution of 30 m. Table I lists the main information about the two AHSI sensors.

B. Two Hyperspectral Datasets

The first hyperspectral dataset is the GF-5 dataset, which consisted of six HSIs acquired by the AHSI aboard the GF-5 satellite. The second hyperspectral dataset, referred to as the

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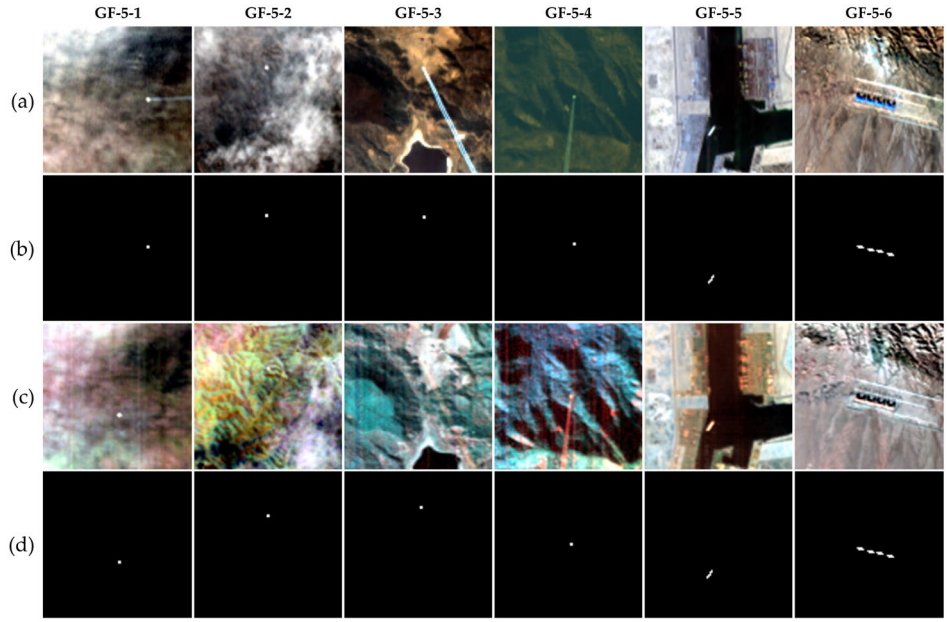


Fig. 4. Six HSIs from the GF-5 dataset: (a) pseudocolor image of the VNIR band, (b) ground truth of the VNIR band, (c) pseudocolor image of the SWIR band, and (d) ground truth of the SWIR band.

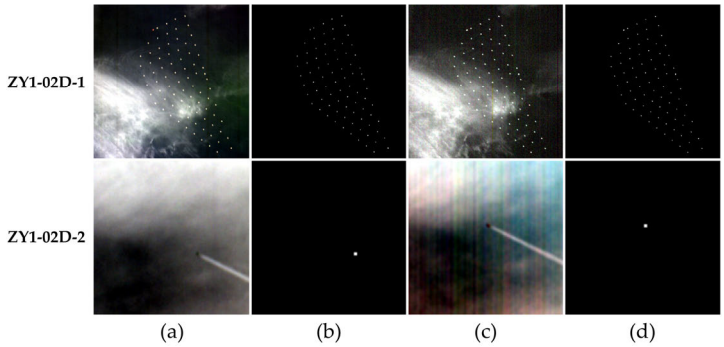


Fig. 5. Two HSIs from the ZY1-02D dataset: (a) pseudocolor image of the VNIR band, (b) ground truth of the VNIR band, (c) pseudocolor image of the SWIR band, and (d) ground truth of the SWIR band.

ZY1-02D dataset, consists of two HSIs acquired by the AHSI aboard the ZY1-02D satellite. These HSIs are representative and encompass various common scenes and anomalies of different sizes. Each HSI in the datasets contains images from two spectral bands: VNIR and SWIR.

In the GF-5 dataset, the first HSI has a cloudy ocean background, and the anomaly is a pixel-level target, denoted as GF-5-1. The second HSI is a cloudy mountainous forest background, and the anomaly is a pixel-level target, denoted as GF-5-2. The third HSI has a cloudless grassland background, and the anomaly is at the pixel level, denoted as GF-5-3. The background of the fourth HSI is a cloudless forest, and the anomaly is also at the pixel level, denoted as GF-5-4. The background of the fifth HSI is a cloudless harbor with an anomaly target comprised of more than 10 pixels, denoted as GF-5-5. The sixth HSI in the dataset contains the background of a cloudless desert with multiple large-scale anomalies and is denoted as GF-5-6. For these six HSIs, we cropped the images around the anomalous targets. After cropping, the size of the VNIR images is $100 \times 100 \times 150$ and that of the SWIR images is $100 \times 100 \times 180$. Fig. 4 shows six cropped HSIs from the GF-5 dataset, with each HSI comprising two parts: a VNIR-band image and an SWIR-band image. We can observe that compared to the VNIR images, the differences between the anomalies and the background are even smaller in the SWIR images, such as GF-5-2 and GF-5-3 In addition, the stripe noise in the SWIR images is more pronounced than that in the VNIR images. Therefore, conducting anomaly detection on both VNIR and SWIR images allows us to verify the performance of the detection methods under different conditions.

In the ZY1-02D dataset, the two HSIs contain two different types of anomalous targets. The first HSI has a cloudy ocean background, and the anomalies are multiple pixel-level targets, denoted as ZY1-02D-1. The second HSI is a cloudy grassland background, and the anomaly is a pixel-level dark target, denoted as ZY1-02D-2. After cropping, the size of the VNIR image of ZY1-02D-1 is $100 \times 100 \times 76$ and that of the SWIR image is $100 \times 100 \times 90$. For ZY1-02D-2, the VNIR image size is $100 \times 100 \times 76$, and the SWIR image size is $100 \times 100 \times 90$. Fig. 5 shows the two cropped HSIs from the ZY1-02D dataset, with each HSI also comprising two parts:

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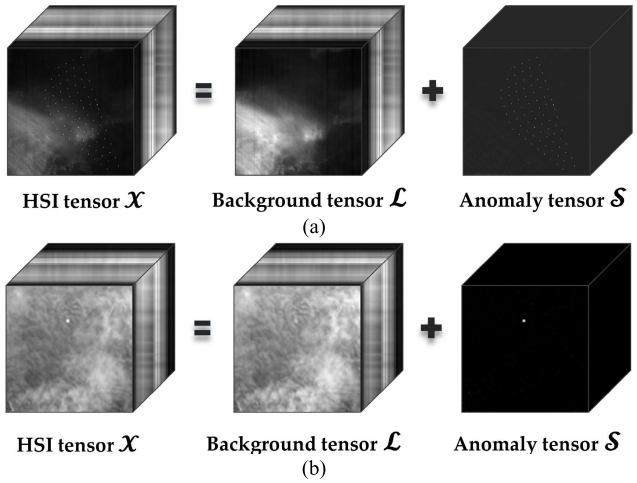


Fig. 6. Illustration of decomposing an HSI tensor into a background tensor and an anomaly tensor: (a) an HSI from the ZY1-02D dataset and (b) an HSI from the GF-5 dataset.

a VNIR band image and an SWIR band image. The HSIs from the two hyperspectral datasets mentioned above are used in subsequent validation experiments.

IV. Proposed Hyperspectral Anomaly Detection Method

A. Problem Formulation

An HSI can be represented as a third-order tensor $\mathcal{X} \in \mathbb{R}^{H\times W\times B}$, where H and W represent the spatial dimensions and B represents the spectral dimension. For hyperspectral anomaly detection, we assume that $\mathcal{X}$ can be decomposed into two parts: a background tensor $\mathcal{L}$ and anomaly tensor $\mathcal{S}$, defined as $\mathcal{X} = \mathcal{L} + \mathcal{S}$. In the optimal decomposition scenario, the background tensor represents the background without anomalous pixels, whereas the anomaly tensor contains only nonzero tube fibers of anomalous pixels, as shown in Fig. 6. The decomposition problem mentioned above is an ill-posed inverse problem [41]. To obtain an optimal solution, the potential (prior) information of the background and anomaly tensors should be utilized fully. Since the background is usually composed of a few types of ground materials, each pixel of the background tensor can be formed by a linear combination of several ground material endmembers. Therefore, the background tensor is an intrinsic low rank in the spatial dimension, corresponding to the frontal slices of the background tensor [47]. In push-broom imaging, the 2-D data acquired by the area-array detector in each frame correspond to a horizontal slice of the HSI tensor. The correlation of adjacent bands in the spectral dimension endows the background tensor with a low-rank property in the spectral dimension, corresponding to the horizontal slices of the background tensor. Therefore, we design a joint low-rank regularization term that simultaneously constrains the low-rank properties of the background tensor in both spatial and spectral dimensions, which is described in the Section IV-B.

Regarding the background tensor, in the spatial dimension, adjacent pixels of the same type of ground material typically exhibit a certain degree of homogeneity, leading to spatial piecewise smoothness. Furthermore, anomalies are less frequent in HSIs, resulting in the tubewise sparsity of the anomaly tensor. Therefore, we establish the following optimization model, which makes full use of the abovementioned intrinsic characteristics of the background and anomaly tensors, and can be represented as

$$\begin{aligned} &\min_{\mathcal{L},\mathcal{S}} \operatorname{rank}_{\text{joint}}(\mathcal{L}) + \mathcal{R}_1(\mathcal{L}) + \mathcal{R}_2(\mathcal{S}) \\ &\text{s.t. } \mathcal{X} = \mathcal{L} + \mathcal{S} \end{aligned} \tag{4}$$

where the $\operatorname{rank}_{\text{joint}}(\mathcal{L})$ term serves as a measure of the joint low-rank of the background tensor in the spatial and spectral dimensions. The regularization term $\mathcal{R}_1(\mathcal{L})$ is employed to enforce the spatial piecewise smoothness of the background tensor, whereas the regularization term $\mathcal{R}_2(\mathcal{S})$ focuses on the tubewise sparsity of the anomaly tensor. In the following, we will provide a detailed analysis and mathematical description of the three terms in the model.

B. Joint Low-Rank Regularization Term

For the background tensor $\mathcal{L} \in \mathbb{R}^{H\times W\times B}$, only imposing a low-rank constraint on the original background tensor using t-SVD does not flexibly characterize the low-rank properties of the spatial and spectral dimensions. Therefore, we design a joint low-rank constraint term to simultaneously constrain the low-rank properties in both spatial and spectral dimensions.

In Fig. 7, the GF-5-1 VNIR image is used as an example to show the different low-rank characteristics exhibited by the background tensor in both spatial and spectral dimensions. For the spatial dimension, the 2-D spatial image of each band corresponds to a frontal slice of the background tensor, as shown in the first row of Fig. 7(b). A background tensor is typically composed of several types of ground materials, and each pixel in the spatial dimension of the background tensor can be formed by a linear combination of several ground material endmembers. This results in the background tensor exhibiting a low-rank characteristic in the frontal slices. As shown in the first row of Fig. 7(c) and (d), we can observe that for each frontal slice of the tensor obtained after the mode-3 fiber Fourier transform of the background tensor, only a few singular values are significantly greater than zero, and the singular values decrease rapidly. This indicates the low-rank characteristic in the spatial dimension.

For the spectral dimension, each 2-D data captured by the area-array detector during the push-broom imaging process, representing spectral information of a single row of ground object, corresponds to a horizontal slice of the background tensor, as illustrated in Fig. 3 and the second row of Fig. 7(b). In the horizontal slices of the background tensor, there is a certain degree of correlation between adjacent bands in the spectral dimension, and the spectral vectors of adjacent similar types of ground materials exhibit similarity. This endows the background tensor with a different low-rank characteristic in the horizontal slices compared to the frontal slices. As shown in the second row of Fig. 7(c) and (d), each horizontal slice of the tensor obtained from the mode-1 fiber Fourier transform of the background tensor has only a few singular values significantly greater than zero. This demonstrates the low-rank

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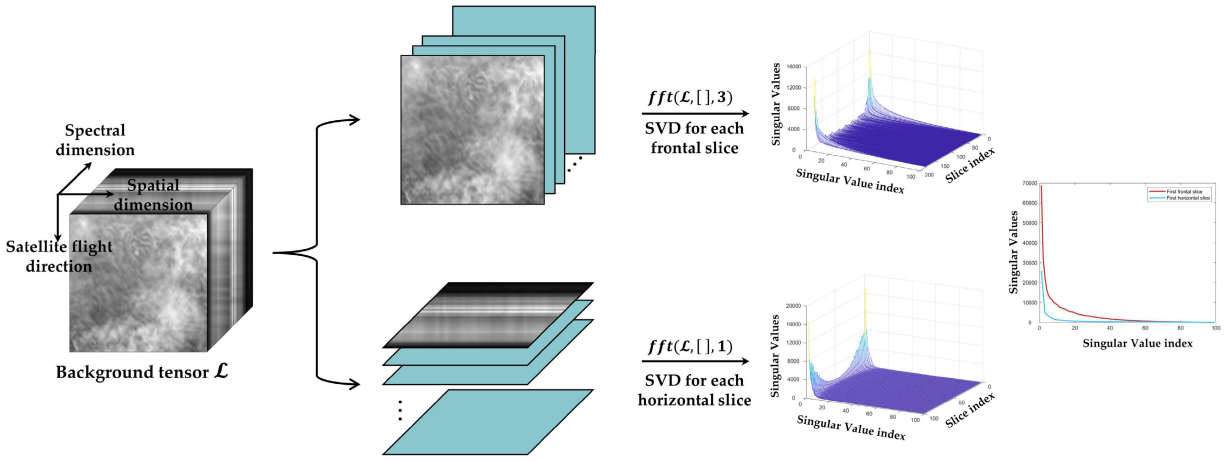


Fig. 7. Illustration of the different low-rank characteristics of the background tensor in the spatial and spectral dimensions: (a) background tensor of the GF-5-1 VNIR image, (b) frontal slices and horizontal slices of the background tensor, (c) singular values on different slices (two-end) of the tensor after performing Fourier transform on mode-1 and mode-3 fibers, and (d) singular values on the first slice of the tensor after performing Fourier transform on mode-1 and mode-3 fibers.

characteristics of the background tensor in the spectral dimension.

Therefore, it is insufficient to apply only spatial low-rank constraints to the background tensor. It is necessary to simultaneously and flexibly apply low-rank constraints to the background tensor in both the spatial and spectral dimensions, for which we have designed the joint low-rank regularization term $\text{rank}_{\text{joint}}(\mathcal{L})$ of the background tensor for the optimization model (4). We use the tensor multirank defined by t-SVD as the measure of tensor rank in the joint low-rank regularization term. The t-SVD characterizes the correlations on the frontal slices of a tensor using SVD and considers the correlations along the mode-3 fibers through circular convolution [48]. By minimizing the tensor nuclear norm, which is the tightest convex relaxation to the ℓ_1-norm of the multirank tensor, constraints can be imposed on the background tensor to exhibit low-rank characteristics in the frontal slices.

However, we can then observe a significant difference in the singular values, as shown in Fig. 7(c) and (d), while the tensor nuclear norm defined in (3) treats different singular values equally. This can lead to an excessive reduction in larger singular values during optimization, resulting in the loss of major components in background tensor estimation [49], [50]. To overcome this drawback, we employ a weighted tensor nuclear norm as the measure of the tensor multirank. The weighted tensor nuclear norm assigns different weights to singular values, enabling better preservation of larger singular values. The corresponding definition is given as follows.

Definition 8 (Weighted Tensor Nuclear Norm [51]): For $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, the t-SVD-based weighted tensor nuclear norm is defined as the sum of the weighted nuclear norm of all the frontal slices of $\overline{\mathcal{A}}$ and is represented as

$$\begin{aligned} \|\mathcal{A}\|_{\boldsymbol{w},\circledast} &= \sum_{i=1}^{n_3} \|\overline{\boldsymbol{A}}^{(i)}\|_{\boldsymbol{w},*} \\ &= \sum_{i=1}^{n_3} \sum_{j=1}^{\min(n_1,n_2)} w_j \cdot \sigma_j\left(\overline{\boldsymbol{A}}^{(i)}\right) \end{aligned} \tag{5}$$

where $\sigma_j(\overline{\boldsymbol{A}}^{(i)})$ denotes the jth largest singular value of $\overline{\boldsymbol{A}}^{(i)}$ and w_j denotes the weight for the singular value of $\sigma_j(\overline{\boldsymbol{A}}^{(i)})$.

For the weight setting of the nuclear norm, larger singular values should be multiplied times smaller weights, whereas smaller singular values should be multiplied times larger weights. Thus, larger singular values shrink less, enabling better preservation of the main components. Therefore, we set the weights inversely proportional to the singular values. Specifically, for the singular value $\sigma_j(\overline{\boldsymbol{A}}^{(i)})$, the weight that we set is given as follows:

$$w_j = \frac{\sqrt{mn}}{\sigma_j\left(\overline{\boldsymbol{A}}^{(i)}\right) + \varepsilon} \tag{6}$$

where m and n are the sizes of the frontal slice $\overline{\boldsymbol{A}}^{(i)}$ and $\varepsilon = 10^{-16}$ is included to avoid dividing by zero.

The weighted tensor nuclear norm $\|\mathcal{L}\|_{\boldsymbol{w},\circledast}$ of the background tensor $\mathcal{L}$ can serve as the low-rank regularization term in the spatial dimension. To utilize the weighted tensor nuclear norm for characterizing the low-rank properties of the background tensor in the spectral dimension simultaneously, we perform a permutation operation on the background tensor $\mathcal{L}$ to obtain the permuted tensor $\mathcal{L}_p$, denoted as $\mathcal{L}_p = \text{permute}(\mathcal{L}, [3, 2, 1])$, as illustrated in Fig. 8. This operation swaps the third and first dimensions of $\mathcal{L}$. Consequently, the frontal slice of $\mathcal{L}_p$ corresponds to the horizontal slice of $\mathcal{L}$. Therefore, the weighted tensor nuclear norm $\|\mathcal{L}_p\|_{\boldsymbol{w},\circledast}$ of the permuted tensor $\mathcal{L}_p$ can be used as the low-rank regularization term for the spectral dimension of the background tensor.

Therefore, we designed a joint low-rank regularization term $\text{rank}_{\text{joint}}(\mathcal{L})$ through the weighted sum of the aforementioned low-rank regularization terms in the spatial and spectral dimensions. Consequently, based on the joint low-rank regularization term, the optimization model (4) can be rewritten as follows:

$$\begin{aligned} &\min_{\mathcal{L},\mathcal{S}} \alpha\|\mathcal{L}\|_{\boldsymbol{w},\circledast} + (1-\alpha)\|\mathcal{L}_p\|_{\boldsymbol{w},\circledast} + \mathcal{R}_1(\mathcal{L}) + \mathcal{R}_2(\mathcal{S}) \\ &\text{s.t. } \mathcal{X} = \mathcal{L} + \mathcal{S} \end{aligned} \tag{7}$$

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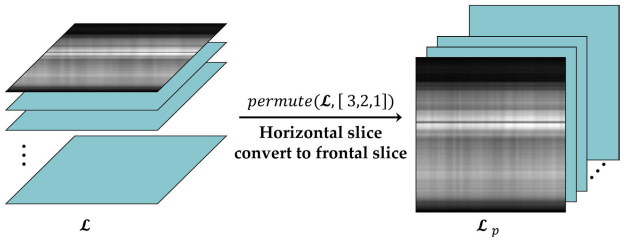


Fig. 8. Illustration of the permutation operation applied to obtain $\mathcal{L}_p$.

where $\|\mathcal{L}\|_{\boldsymbol{w},\circledast}$ is used to constrain the low rank characteristic of the background tensor $\mathcal{L}$ in the spatial dimension, $\|\mathcal{L}_p\|_{\boldsymbol{w},\circledast}$ is used to constrain the low rank characteristic of the background tensor $\mathcal{L}$ in the spectral dimension. In addition, α is the balance parameter employed to balance the degree of low-rankness in the two dimensions.

C. Linear TV-Norm Regularization Term

For the background tensor, adjacent pixels are more likely to be of the same type of surface material, leading to a certain degree of spatial homogeneity in these pixels [52]. This property of spatial homogeneity in neighboring pixels imparts the background tensor with spatial piecewise smoothness. Therefore, applying a piecewise smoothness constraint to the background tensor helps to better separate anomalies from the background, enabling the anomaly tensor to more effectively capture anomalous pixels, thereby enhancing detection performance.

The TV model was proposed in [53] and has been widely applied in image processing. By utilizing TV regularization, a spatial piecewise smoothing constraint can be imposed on the background tensor. Given a matrix $\boldsymbol{X} \in \mathbb{R}^{H\times W}$, the TV-norm is defined as follows:

$$\|\boldsymbol{X}\|_{TV} = \sum_{i=1}^{H}\sum_{j=1}^{W}\sqrt{\boldsymbol{X}_v(i,j)^2 + \boldsymbol{X}_h(i,j)^2} \tag{8}$$

$$\boldsymbol{X}_v(i,j) = \begin{cases} \boldsymbol{X}(i+1,j) - \boldsymbol{X}(i,j), & \text{if } i < H \\ 0, & \text{if } i = H \end{cases} \tag{9}$$

$$\boldsymbol{X}_h(i,j) = \begin{cases} \boldsymbol{X}(i,j+1) - \boldsymbol{X}(i,j), & \text{if } j < W \\ 0, & \text{if } j = W. \end{cases} \tag{10}$$

Although the problem of minimizing the TV-norm is convex and can be solved using gradient methods, it does not have a closed-form solution [30]. Therefore, we employed the linear TV-norm proposed in [30] instead of the original TV-norm to enforce spatial piecewise smoothness. The linear TV-norm is defined as

$$\|\boldsymbol{X}\|_{LTV} = \sum_{i=1}^{H}\sum_{j=1}^{W}\boldsymbol{X}_v(i,j)^2 + \boldsymbol{X}_h(i,j)^2. \tag{11}$$

We extend this linear TV-norm to an HSI in a band-by-band manner and use it to enforce a spatial piecewise smoothness constraint on the background tensor $\mathcal{L}$. Therefore, based on the linear TV-norm, the regularization term $\mathcal{R}_1(\mathcal{L})$ in (4) is defined as follows:

$$\mathcal{R}_1(\mathcal{L}) = \beta\left(\|\boldsymbol{D}_H\boldsymbol{L}_{(1)}\|_F^2 + \|\boldsymbol{D}_W\boldsymbol{L}_{(2)}\|_F^2\right) \tag{12}$$

where β is regularized parameter, $\boldsymbol{L}_{(1)} \in \mathbb{R}^{H\times WB}$ is obtained by $\boldsymbol{L}_{(1)} = \text{unfold}_{\,1}(\mathcal{L})$, $\boldsymbol{L}_{(2)} \in \mathbb{R}^{W\times HB}$ is obtained by $\boldsymbol{L}_{(2)} = \text{unfold}_{\,2}(\mathcal{L})$, and $\boldsymbol{D}_H \in \mathbb{R}^{(H-1)\times H}$ and $\boldsymbol{D}_W \in \mathbb{R}^{(W-1)\times W}$ are two dimensional finite-difference operators, defined as follows:

$$\boldsymbol{D}_H(i,j) = \begin{cases} 1, & \text{if } j = i \\ -1, & \text{if } j = i+1 \\ 0, & \text{others} \end{cases} \tag{13}$$

$$\boldsymbol{D}_W(i,j) = \begin{cases} 1, & \text{if } j = i \\ -1, & \text{if } j = i+1 \\ 0, & \text{others.} \end{cases} \tag{14}$$

D. $\ell_{1,1,2}$-Norm Regularization Term

For satellite-borne HSIs, anomalies typically occur at the pixel level and occupy only a small portion of the spatial domain. Therefore, an optimal anomaly tensor containing only anomalous pixels exhibits sparse characteristics. Furthermore, the decomposition process should preserve the original physical characteristics of anomalies. In the anomaly tensor, the anomalies should retain their full spectral information, which is to say that the tube fibers associated with these anomalies should be intact and nonzero throughout. Therefore, the characteristics of an anomalous tensor should be tubewise sparse. Similar to the utilization of the $\ell_{2,1}$-norm to enforce column sparsity in the matrix, as proposed in [54], we employed the $\ell_{1,1,2}$-norm [36] to constrain the tubewise sparse characteristic of the anomaly tensor. Based on the $\ell_{1,1,2}$-norm, the regularization term $\mathcal{R}_2(\mathcal{S})$ in model (4) is defined as follows:

$$\mathcal{R}_2(\mathcal{S}) = \lambda\|\mathcal{S}\|_{1,1,2} = \lambda\sum_{i,j}\|\mathcal{S}(i,j,:)\|_F \tag{15}$$

where λ is a regularization parameter.

Based on the aforementioned analysis, by combining (7), (12), and (15), we obtain the complete anomaly detection model, as follows:

$$\begin{aligned} \min_{\mathcal{L},\mathcal{S}}\ & \alpha\|\mathcal{L}\|_{\boldsymbol{w},\circledast} + (1-\alpha)\|\mathcal{L}_p\|_{\boldsymbol{w},\circledast} \\ & + \beta\left(\|\boldsymbol{D}_H\boldsymbol{L}_{(1)}\|_F^2 + \|\boldsymbol{D}_W\boldsymbol{L}_{(2)}\|_F^2\right) + \lambda\|\mathcal{S}\|_{1,1,2} \\ \text{s.t.}\ & \mathcal{X} = \mathcal{L} + \mathcal{S}. \end{aligned} \tag{16}$$

The first two terms of the objective function in model (16) constitute the joint low-rank regularization term, designed to flexibly constrain the low-rank properties of the background tensor in both the spatial and spectral dimensions. The third and fourth items are linear TV-norm regularization, used to impose spatial piecewise smoothness constraints on the background tensor. The final item is the $\ell_{1,1,2}$-norm regularization, used to constrain the tubewise sparsity of the anomaly tensor. The parameters α, β, and λ are used to balance the effects of each term, and we analyze the settings of these parameters in Section V-B. Next, we provide a detailed introduction of the solving algorithm designed for this model.

E. ADMM-Based Solving Algorithm

The ADMM is an optimization algorithm proposed in [53]. It tackles optimization problems with coupled variables and

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constraints by iteratively updating the variables and Lagrange multipliers. The ADMM has found wide applications in distributed optimization, image processing, and machine learning. In this study, we used the ADMM to solve the optimization problem presented in (16).

Firstly, we introduce the auxiliary variables $\mathcal{Z}^1$, $\mathcal{Z}^2$, $\mathcal{Z}^3$, and $\mathcal{Z}^4$ to separate the variables in the objective function of (16). This model can be reformulated as follows:

$$\begin{aligned}\min_{\mathcal{L},\mathcal{S}} &\alpha\|\mathcal{Z}^1\|_{w,\circledast} + (1-\alpha)\|\mathcal{Z}_p^2\|_{w,\circledast} \\ &+ \beta\left(\|\boldsymbol{D}_H \boldsymbol{Z}_{(1)}^3\|_F^2 + \|\boldsymbol{D}_W \boldsymbol{Z}_{(2)}^4\|_F^2\right) + \lambda\|\mathcal{S}\|_{1,1,2} \\ \text{s.t. } &\mathcal{X} = \mathcal{L} + \mathcal{S}, \mathcal{Z}^1 = \mathcal{L}, \mathcal{Z}^2 = \mathcal{L}, \mathcal{Z}^3 = \mathcal{L}, \mathcal{Z}^4 = \mathcal{L}.\end{aligned} \tag{17}$$

We then obtain the augmented Lagrangian function of (17) as follows:

$$\begin{aligned}&\mathrm{Lag}\left(\mathcal{L}, \mathcal{S}, \mathcal{Z}^1, \mathcal{Z}^2, \mathcal{Z}^3, \mathcal{Z}^4, \mathcal{M}^1, \mathcal{M}^2, \mathcal{M}^3, \mathcal{M}^4, \mathcal{P}\right) \\ &= \alpha\|\mathcal{Z}^1\|_{w,\circledast} + (1-\alpha)\|\mathcal{Z}_p^2\|_{w,\circledast} \\ &\quad + \beta\left(\|\boldsymbol{D}_H \boldsymbol{Z}_{(1)}^3\|_F^2 + \|\boldsymbol{D}_W \boldsymbol{Z}_{(2)}^4\|_F^2\right) + \lambda\|\mathcal{S}\|_{1,1,2} \\ &\quad + \frac{\mu}{2}\|\mathcal{L} - \mathcal{Z}^1 + \frac{\mathcal{M}^1}{\mu}\|_F^2 + \frac{\mu}{2}\|\mathcal{L} - \mathcal{Z}^2 + \frac{\mathcal{M}^2}{\mu}\|_F^2 \\ &\quad + \frac{\mu}{2}\|\mathcal{L} - \mathcal{Z}^3 + \frac{\mathcal{M}^3}{\mu}\|_F^2 + \frac{\mu}{2}\|\mathcal{L} - \mathcal{Z}^4 + \frac{\mathcal{M}^4}{\mu}\|_F^2 \\ &\quad + \frac{\mu}{2}\|\mathcal{X} - \mathcal{L} - \mathcal{S} + \frac{\mathcal{P}}{\mu}\|_F^2\end{aligned} \tag{18}$$

where $\mathcal{M}^1$, $\mathcal{M}^2$, $\mathcal{M}^3$, $\mathcal{M}^4$, and $\mathcal{P}$ are the Lagrange multipliers and μ is the positive penalty parameter.

Within the ADMM framework, (18) can be approximately solved by alternatively updating the variables while fixing the others. Specifically, for the $k+1$th iteration, the variables are updated by sequentially solving the following subproblems.

1) Update $\mathcal{Z}^1$: Fixing the variables other than $\mathcal{Z}^1$, the resulting subproblem regarding $\mathcal{Z}^1$ is given as follows:

$$\mathcal{Z}_{k+1}^1 = \arg\min_{\mathcal{Z}^1} \frac{\alpha}{\mu_k}\|\mathcal{Z}_k^1\|_{w,\circledast} + \frac{1}{2}\left(\|\mathcal{Z}_k^1 - \left(\mathcal{L}_k + \frac{\mathcal{M}_k^1}{\mu_k}\right)\|_F^2\right). \tag{19}$$

The circular convolution operation on the tube fibers of the t-SVD can be viewed as the SVD of the frontal slice of the tensor in the Fourier domain. Therefore, similar to using the singular value thresholding (SVT) method [54] to solve the optimization problems associated with the matrix nuclear norm, we employed the weighted tensor SVT (WTSVT) method [51] to solve the problem in (19). The details of the WTSVT are presented in Algorithm 1.

Next, we provide a theorem for utilizing the WTSVT method to solve the weighted tensor nuclear norm minimization problem.

Theorem 2: (weighted tensor muclear norm minimization [57]): For a three-order tensor $\mathcal{Y} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, its t-SVD factorization is denoted as $\mathcal{Y} = \mathcal{U} * \mathcal{S} * \mathcal{V}^T$. For the weighted tensor nuclear norm minimization problem,

$$\mathcal{X}^* = \arg\min_{\mathcal{X}} \alpha\|\mathcal{X}\|_{w,\circledast} + \frac{1}{2}\left(\|\mathcal{X} - \mathcal{Y}\|_F^2\right). \tag{20}$$

Algorithm 1 WTSVT Method

Input: $\mathcal{Y} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, $\boldsymbol{w}$, τ

1: $\overline{\mathcal{Y}} = fft(\mathcal{Y}, [], 3)$
2: **for** $i = 1$ **to** n_3 **do**
3: $[\boldsymbol{U}, \boldsymbol{S}, \boldsymbol{V}] = \mathrm{SVD}(\overline{\mathcal{Y}}(:,:,i))$
4: $\boldsymbol{s} = \max(diag(\boldsymbol{S}) - \boldsymbol{w} * \tau, 0)$
5: $\boldsymbol{S} = diag(\boldsymbol{s})$
6: $\overline{\mathcal{U}}(:,:,i) = \boldsymbol{U}, \overline{\mathcal{S}}(:,:,i) = \boldsymbol{S}, \overline{\mathcal{V}}(:,:,i) = \boldsymbol{V}$
7: end for
8: $\mathcal{U} = ifft(\overline{\mathcal{U}}, [], 3), \mathcal{S} = ifft(\overline{\mathcal{S}}, [], 3), \mathcal{V} = ifft(\overline{\mathcal{V}}, [], 3)$
9: $\mathcal{D}_{\tau,\boldsymbol{w}}(\mathcal{Y}) = \mathcal{U} * \mathcal{S} * \mathcal{V}$

Output: $\mathcal{D}_{\tau,\boldsymbol{w}}(\mathcal{Y}) \in \mathbb{R}^{n_1 \times n_2 \times n_3}$

The optimal solution can be obtained using the WTSVT operator, denoted as

$$\mathcal{X}^* = \mathcal{D}_{\alpha,\boldsymbol{w}}(\mathcal{Y}) \tag{21}$$

Therefore, the solution of (19) can be obtained as follows:

$$\mathcal{Z}_{k+1}^1 = \mathcal{D}_{\frac{\alpha}{\mu_k},\boldsymbol{w}}(\mathcal{T}_k^1) \tag{22}$$

where $\mathcal{D}_{\frac{\alpha}{\mu_k},\boldsymbol{w}}(\cdot)$ is the WTSVT operator and $\mathcal{T}_k^1 = \mathcal{L}_k + \frac{\mathcal{M}_k^1}{\mu_k}$.

2) Update $\mathcal{Z}^2$: Fixing the variables other than $\mathcal{Z}^2$, the resulting sub-problem regarding $\mathcal{Z}^2$ is as follows:

$$\mathcal{Z}_{k+1}^2 = \arg\min_{\mathcal{Z}^2} \frac{1-\alpha}{\mu_k}\|\mathcal{Z}_{k,p}^2\|_{w,\circledast} + \frac{1}{2}\left(\|\mathcal{Z}_k^2 - \left(\mathcal{L}_k + \frac{\mathcal{M}_k^2}{\mu_k}\right)\|_F^2\right). \tag{23}$$

Similar to the process of updating $\mathcal{Z}^1$, the optimal solution for this sub-problem can be obtained using the WTSVT operator and tensor permutation. Firstly, we define $\mathcal{T}_k^2$ as follows:

$$\mathcal{T}_k^2 = \mathcal{L}_k + \frac{\mathcal{M}_k^2}{\mu_k}. \tag{24}$$

Then, we perform tensor permutation operation on $\mathcal{T}_k^2$, as follows:

$$\mathcal{T}_{k,p}^2 = \mathrm{permute}\left(\mathcal{T}_k^2, [3, 2, 1]\right). \tag{25}$$

Finally, utilizing the WTSVT operator, we obtain the solution to (23) as follows:

$$\mathcal{Z}_{k+1,p}^2 = \mathcal{D}_{\frac{1-\alpha}{\mu_k},\boldsymbol{w}}(\mathcal{T}_{k,p}^2), \tag{26}$$

$$\mathcal{Z}_{k+1}^2 = \mathrm{permute}\left(\mathcal{Z}_{k+1,p}^2, [3, 2, 1]\right). \tag{27}$$

3) Update $\mathcal{Z}^3$: Fixing the variables other than $\mathcal{Z}^3$, the resulting sub-problem regarding $\mathcal{Z}^3$ is as follows:

$$\mathcal{Z}_{k+1}^3 = \arg\min_{\mathcal{Z}^3} \frac{\beta}{\mu_k}\|\boldsymbol{D}_H \boldsymbol{Z}_{k,(1)}^3\|_F^2 + \frac{1}{2}\left(\|\mathcal{Z}_k^3 - \left(\mathcal{L}_k + \frac{\mathcal{M}_k^3}{\mu_k}\right)\|_F^2\right). \tag{28}$$

Let $\mathcal{T}_k^3 = \mathcal{L}_k + \frac{\mathcal{M}_k^3}{\mu_k}$, and perform mode-1 matrixization on it, denoted as

$$\boldsymbol{T}_{k,(1)}^3 = \text{ unfold }_1\left(\mathcal{T}_k^3\right). \tag{29}$$

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Then, the closed-form solution to (28) is given by

$$\boldsymbol{Z}^3_{k+1,(1)} = \frac{\beta}{\mu_k}\left(\boldsymbol{D}_H^T\boldsymbol{D}_H + \frac{\beta}{\mu_k}\boldsymbol{I}_H\right)^{-1}\boldsymbol{T}^3_{k,(1)} \tag{30}$$

$$\mathcal{Z}^3_{k+1} = \text{fold}_1\left(\boldsymbol{Z}^3_{k+1,(1)}\right) \tag{31}$$

where $\boldsymbol{I}_H \in \mathbb{R}^{H\times H}$ is an identity matrix.

4) Update $\mathcal{Z}^4$: Fixing the variables other than $\mathcal{Z}^4$, the resulting sub-problem regarding $\mathcal{Z}^4$ is as follows:

$$\mathcal{Z}^4_{k+1} = \arg\min_{\mathcal{Z}^4} \frac{\beta}{\mu_k}\|\boldsymbol{D}_W\boldsymbol{Z}^4_{k,(2)}\|_F^2 + \frac{1}{2}\left(\|\mathcal{Z}^4_k - \left(\mathcal{L}_k + \frac{\mathcal{M}^4_k}{\mu_k}\right)\|_F^2\right). \tag{32}$$

Similar to the case of $\mathcal{Z}^3$, let $\mathcal{T}^4_k = \mathcal{L}_k + \frac{\mathcal{M}^4_k}{\mu_k}$, and obtain the closed-form solution to (32) through the following formulas:

$$\boldsymbol{T}^4_{k,(2)} = \text{unfold}_2\left(\mathcal{T}^4_k\right) \tag{33}$$

$$\boldsymbol{Z}^4_{k+1,(2)} = \frac{\beta}{\mu_k}\left(\boldsymbol{D}_W^T\boldsymbol{D}_W + \frac{\beta}{\mu_k}\boldsymbol{I}_W\right)^{-1}\boldsymbol{T}^4_{k,(2)} \tag{34}$$

$$\mathcal{Z}^4_{k+1} = \text{fold}_2\left(\boldsymbol{Z}^4_{k+1,(2)}\right) \tag{35}$$

where $\boldsymbol{I}_W \in \mathbb{R}^{W\times W}$ is an identity matrix.

5) Update $\mathcal{L}$: Fixing the other variables, update $\mathcal{L}$ by solving the following sub-problem:

$$\begin{aligned}\mathcal{L}_{k+1} = \arg\min_{\mathcal{L}} \frac{\mu_k}{2}\Bigg(&\|\mathcal{L}_k - \left(\mathcal{X} - \mathcal{S}_k + \frac{\mathcal{P}_k}{\mu_k}\right)\|_F^2 \\ &+ \|\mathcal{L}_k - \left(\mathcal{Z}^1_{k+1} - \frac{\mathcal{M}^1_k}{\mu_k}\right)\|_F^2 + \|\mathcal{L}_k - \left(\mathcal{Z}^2_{k+1} - \frac{\mathcal{M}^2_k}{\mu_k}\right)\|_F^2 \\ &+ \|\mathcal{L}_k - \left(\mathcal{Z}^3_{k+1} - \frac{\mathcal{M}^3_k}{\mu_k}\right)\|_F^2 + \|\mathcal{L}_k - \left(\mathcal{Z}^4_{k+1} - \frac{\mathcal{M}^4_k}{\mu_k}\right)\|_F^2\Bigg).\end{aligned} \tag{36}$$

Then, the closed-form solution to (36) is given by

$$\begin{aligned}\mathcal{L}_{k+1} = \frac{1}{5}\Bigg[&\left(\mathcal{Z}^1_{k+1} - \frac{\mathcal{M}^1_k}{\mu_k}\right) + \left(\mathcal{Z}^2_{k+1} - \frac{\mathcal{M}^2_k}{\mu_k}\right) + \left(\mathcal{Z}^3_{k+1} - \frac{\mathcal{M}^3_k}{\mu_k}\right) \\ &+ \left(\mathcal{Z}^4_{k+1} - \frac{\mathcal{M}^4_k}{\mu_k}\right) + \left(\mathcal{X} - \mathcal{S}_k + \frac{\mathcal{P}_k}{\mu_k}\right)\Bigg].\end{aligned} \tag{37}$$

6) Update $\mathcal{S}$: Fixing the other variables, the sub-problem of $\mathcal{S}$ can be written as

$$\begin{aligned}\mathcal{S}_{k+1} = \arg\min_{\mathcal{S}} &\lambda\|\mathcal{S}_k\|_{1,1,2} \\ &+ \frac{\mu_k}{2}\left(\|\mathcal{S}_k - \left(\mathcal{X} - \mathcal{L}_{k+1} + \frac{\mathcal{P}_k}{\mu_k}\right)\|_F^2\right).\end{aligned} \tag{38}$$

For the $\ell_{1,1,2}$-norm minimization sub-problem of (38), its closed-form solution can be obtained as follows [42]:

$$\mathcal{S}_{k+1}(i,j,:) = \max\left(1 - \frac{\lambda}{\mu_k\|\mathcal{T}^5_k(i,j,:)\|}, 0\right)\mathcal{T}^5_k(i,j,:) \tag{39}$$

where $\mathcal{T}^5_k = \mathcal{X} - \mathcal{L}_{k+1} + \frac{\mathcal{P}_k}{\mu_{k^*}}$.

7) Update Lagrange multipliers $\mathcal{M}^1, \mathcal{M}^2, \mathcal{M}^3, \mathcal{M}^4$, and $\mathcal{P}$ and penalty parameter μ: The multipliers and penalty parameter used in optimization process can be updated using the following formulas:

$$\mathcal{M}^1_{k+1} = \mathcal{M}^1_k + \mu_k\left(\mathcal{L}_{k+1} - \mathcal{Z}^1_{k+1}\right) \tag{40}$$

$$\mathcal{M}^2_{k+1} = \mathcal{M}^2_k + \mu_k\left(\mathcal{L}_{k+1} - \mathcal{Z}^2_{k+1}\right) \tag{41}$$

$$\mathcal{M}^3_{k+1} = \mathcal{M}^3_k + \mu_k\left(\mathcal{L}_{k+1} - \mathcal{Z}^3_{k+1}\right) \tag{42}$$

$$\mathcal{M}^4_{k+1} = \mathcal{M}^4_k + \mu_k\left(\mathcal{L}_{k+1} - \mathcal{Z}^4_{k+1}\right) \tag{43}$$

$$\mathcal{P}_{k+1} = \mathcal{P}_k + \mu_k\left(\mathcal{X} - \mathcal{L}_{k+1} - \mathcal{S}_{k+1}\right) \tag{44}$$

$$\mu_{k+1} = \min(\rho\mu_k, \mu_{\max}) \tag{45}$$

where $\rho = 1.1$ is used to is control the increase of μ.

After the iterative optimization process, we obtain the optimal background tensor $\mathcal{L}^*$ and anomaly tensor $\mathcal{S}^*$. Then, based on the optimal anomaly tensor $\mathcal{S}^*$, we obtain the anomaly detection map $\boldsymbol{R}$ as follows:

$$\boldsymbol{R}(i,j) = \|\mathcal{S}^*(i,j,:)\|_F. \tag{46}$$

We summarize the whole procedure described above in Algorithm 2.

F. Convergence Analysis

Then, we provide a convergence analysis of Algorithm 2 for optimizing the problem (17). For the objective function in (17), the last three function terms are convex functions. When the objective function is convex, the convergence of ADMM is fully proven in [55]. However, the first two terms of the objective function are weighted tensor nuclear norms, which are nonconvex functions, making it theoretically difficult to provide a convergence analysis of Algorithm 2. Nevertheless, in [51], there is a similar case that can be as a reference for convergence analysis. The authors also apply the weighted tensor nuclear norm to tensor completion problems and use ADMM to solve the problem. Furthermore, they prove the Karush–Kuhn–Tucker (KKT) conditions of the algorithm.

Next, taking the real HSI GF-5-1 as an example, we provide an empirical convergence analysis. Let $\mathcal{X}$ be the input HSI tensor and $\{\mathcal{L}_k, \mathcal{S}_k, \mathcal{Z}^1_k, \mathcal{Z}^2_k, \mathcal{Z}^3_k, \mathcal{Z}^4_k\}$ be the tensor sequence generated the k-th iteration by Algorithm 2. We define three residuals $\boldsymbol{r}_L, \boldsymbol{r}_S$ and $\boldsymbol{r}_X$, as follows.

$$\boldsymbol{r}_L(k) = \|\mathcal{L}_{k+1} - \mathcal{L}_k\|_F \tag{47}$$

$$\boldsymbol{r}_S(k) = \|\boldsymbol{S}_{k+1} - \boldsymbol{S}_k\|_F \tag{48}$$

$$\boldsymbol{r}_X(k) = \|\mathcal{X} - \mathcal{L}_k - \mathcal{S}_k\|_F. \tag{49}$$

According to the convergence conditions defined in [51] for the ADMM, the aforementioned $\boldsymbol{r}_L$ and $\boldsymbol{r}_S$ represent dual residuals, which measure the difference between the current iteration's solution for dual variables and the solution from the previous iteration, and $\boldsymbol{r}_X$ represents the primal residual, which measures the difference between the current iteration's solution and the original constraints of the problem. If our solving algorithm converges, all three residuals will converge to zero during the iteration process. We use the GF-5-1 HSI as the input tensor for Algorithm 2. The number of iterations is set at 100. The convergence curves of the aforementioned three

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residuals $\boldsymbol{r}_L$, $\boldsymbol{r}_S$, and $\boldsymbol{r}_X$ are depicted in Fig. 9. The curves for all three residuals show a downward trend and converge to zero within several tens of iterations, demonstrating the convergence of Algorithm 2.

Algorithm 2 Proposed TDTLR Anomaly Detection Method

Input: HSI tensor $\boldsymbol{\mathcal{X}} \in \mathbb{R}^{\mathrm{H}\times\mathrm{W}\times\mathrm{B}}$, parameters α, β, λ
Initialize: $\boldsymbol{\mathcal{L}}_0, \boldsymbol{\mathcal{S}}_0, \boldsymbol{\mathcal{Z}}_0^1, \boldsymbol{\mathcal{Z}}_0^2, \boldsymbol{\mathcal{Z}}_0^3, \boldsymbol{\mathcal{Z}}_0^4, \boldsymbol{\mathcal{M}}_0^1, \boldsymbol{\mathcal{M}}_0^2, \boldsymbol{\mathcal{M}}_0^3, \boldsymbol{\mathcal{M}}_0^4, \boldsymbol{\mathcal{P}}_0$ initialized to $0, \mu_0 = 10^{-2}, \rho = 1.1, \mu_{max} = 10^{10}, \xi = 10^{-6}, k = 0, maxIter = 100$.
1: while not converged do
2: Update $\boldsymbol{\mathcal{Z}}_{k+1}^1$ by using (22).
3: Update $\boldsymbol{\mathcal{Z}}_{k+1}^2$ by applying (24)-(27).
4: Update $\boldsymbol{\mathcal{Z}}_{k+1}^3$ by utilizing (29)-(31).
5: Update $\boldsymbol{\mathcal{Z}}_{k+1}^4$ by employing (33)-(35).
6: Update $\boldsymbol{\mathcal{L}}_{k+1}$ by using (37).
7: Update $\boldsymbol{\mathcal{S}}_{k+1}$ by utilizing (39).
8: Update $\boldsymbol{\mathcal{M}}_{k+1}^1, \boldsymbol{\mathcal{M}}_{k+1}^2, \boldsymbol{\mathcal{M}}_{k+1}^3, \boldsymbol{\mathcal{M}}_{k+1}^4, \boldsymbol{\mathcal{P}}_{k+1}$ and μ_{k+1} by applying (40)-(45).
9: check convergence:
$(\|\boldsymbol{\mathcal{L}}_{k+1} - \boldsymbol{\mathcal{L}}_k\|_F \le \xi$ and $\|\boldsymbol{\mathcal{S}}_{k+1} - \boldsymbol{\mathcal{S}}_k\|_F \le \xi$ *and* $\|\boldsymbol{\mathcal{X}} - \boldsymbol{\mathcal{L}}_{k+1} - \boldsymbol{\mathcal{S}}_{k+1}\|_F \le \xi$) *or* $k > maxiter$
10: end while
11: Obtain the anomaly detection map by employing (46).
Output: $\boldsymbol{\mathcal{L}}^*, \boldsymbol{\mathcal{S}}^*, \boldsymbol{R}$.

V. Experiments

A. Evaluation Metrics

Several commonly used metrics are employed to evaluate the proposed method. The first metric is the detection map, which provides a visual representation of the detection results [58]. The second metric is the receiver operating characteristic (ROC) curve [59] with the false alarm rate (FAR) on the horizontal axis and the probability of detection (PD) on the vertical axis. The closer the ROC curve is to the upper left corner, the better the detection performance. The third metric is the area under the ROC curve (AUC). The AUC is a quantitative analysis metric for the ROC curve, with values closer to 1 indicating better detection performance. The fourth metric is a separability map. The detection values for both the background and anomalies distributed within the [10%, 90%] range are represented by the boxes.

B. Parameter Analysis

Our proposed method includes three parameters that need to be adjusted: the balance parameter α of the joint low-rank constraint in two dimensions, the regularization parameter β of spatial piecewise smoothness, and the regularization parameter τ of tubewise sparsity. We use one HSI from each of the two datasets as an example (GF-5-6 and ZY1-02D-1) to demonstrate the impacts of the three parameters on the detection performance, evaluated using the AUC metric, as shown in Fig. 10. When obtaining results for the impact of one parameter on detection performance, this parameter is varied while keeping the other two parameters fixed.

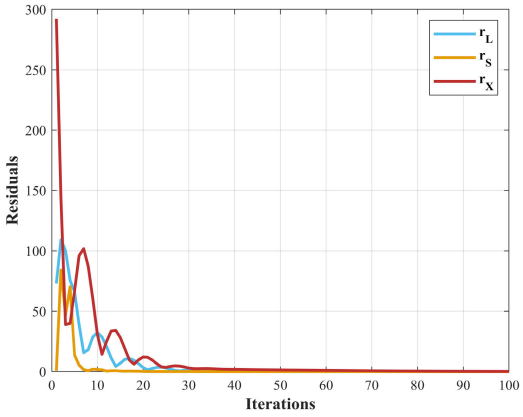


Fig. 9. Convergence curves of the three residuals $\boldsymbol{r}_L$, $\boldsymbol{r}_S$, and $\boldsymbol{r}_X$.

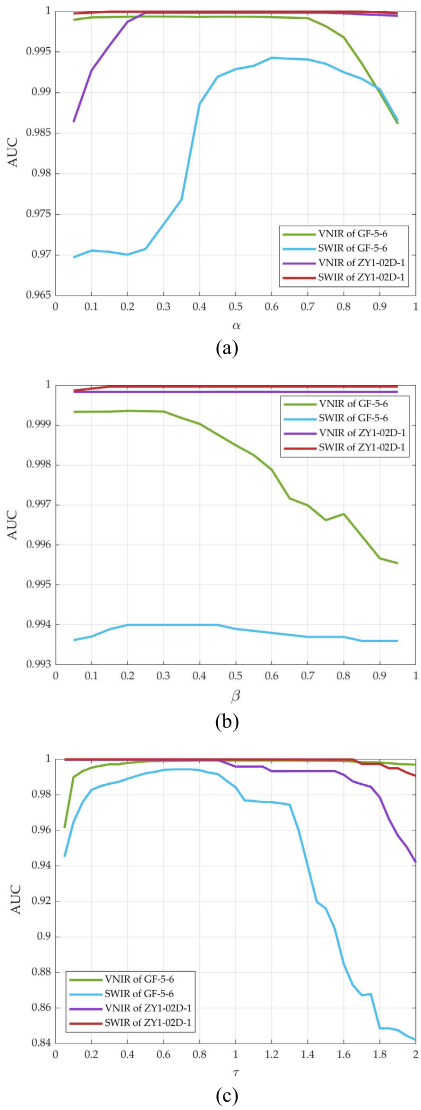


Fig. 10. Values of AUC with different parameter settings: (a) α, (b) β, and (c) τ.

We varied α in increments of 0.05 within the range [0.05, 1]. Fig. 10(a) demonstrates that both excessively large and excessively small values of α lead to deteriorated detection results, decreasing the AUC. When α is set to 1, the

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low-rank constraint term for the background tensor in the spectral dimension becomes 0. In this case, only the spatial-dimensional low-rank constraint is applied to the background tensor. As shown in Fig. 10(a), optimal detection performance cannot be achieved under this condition. On the other hand, when α is set to 0, the spatial-dimensional low-rank constraint term on the background tensor becomes 0, which means that the background tensor is subject only to a low-rank constraint in the spectral dimension. Similarly, as depicted in Fig. 10(a), optimal detection performance is not attainable in this scenario as well. When α takes an appropriate value between 0 and 1, optimal detection performance can be achieved. This is because setting an appropriate value for α ensures that the low-rank characteristics of the background tensor are well constrained in both spatial and spectral dimensions simultaneously, thereby enhancing the detection performance. This fully verifies that the joint low-rank regularization of the background tensor can improve the performance of anomaly detection based on the low-rank decomposition model and promote the application of the low-rank decomposition model in anomaly detection.

As shown in Table I, HSIs in the ZY1-02D dataset have nearly half the number of spectral bands compared to HSIs in the GF-5 dataset. As a result, the rank of HSIs in the spectral dimension is lower for ZY1-02D compared to GF-5. Therefore, a stronger low-rank constraint needs to be applied to the spectral dimension of the ZY1-02D HSIs compared to the GF-5 HSIs. A smaller value of α indicates a weaker low-rank constraint in the spatial dimension and a stronger low-rank constraint in the spectral dimension. Therefore, in our experiments, we set $\alpha = 0.6$ for the GF-5 dataset and set $\alpha = 0.3$ for the ZY1-02D dataset, achieving satisfactory results.

As β increases, the spatial piecewise smoothness of the background tensor becomes stronger, which increases the likelihood of the anomaly tensor containing anomaly pixels, thus improving the PD. However, if the value of β is too large, it can lead to a higher FAR and decrease the detection performance. Fig. 10(b) illustrates the impact of β on the detection performance. We also varied β in increments of 0.05 within the range [0.05, 1]. It can be observed that good detection performance can be achieved for β values ranging from 0.2 to 0.3. Therefore, in our experiments, we set $\beta = 0.3$ for all HSIs.

The role of τ is to control the sparsity level of the anomaly tensor. The larger the value of τ, the fewer nonzero tube fibers are included in the anomaly tensor, which means that the anomaly tensor is sparser. We varied τ in increments of 0.05 within the range [0.05, 2] to investigate the impact of τ on the detection performance, as shown in Fig. 10(c). When τ is too large, the anomaly tensor can become excessively sparse, leading to the possibility of missing anomalies during the detection process. Observably, the detection performs well when τ falls within the range from 0.4 to 1. Therefore, in our experiments, we set τ to 0.5 for HSIs with a larger number of pixels corresponding to the interested anomalies, such as GF-5-5, GF-5-6, and ZY1-02D-1. For other HSIs with fewer pixels corresponding to the interested anomalies, we set τ to 1.

C. Comparison Methods and Detection Performance

To provide a comprehensive validation of the detection performance of our proposed method, we compare it with seven different hyperspectral anomaly detection methods, including the classic statistics-based RX method [11], the low-rank representation-based LRASR [28] and GTVLRR [29] methods, the matrix decomposition-based LSMAD method [27], the deep-learning-based robust graph autoencoder (RGAE) method [23], the spatial–spectral joint-based visual attention and background subtraction (VABS) method [60], and the tensor decomposition-based PTA method [30]. In our experiments, the parameters for these methods are set according to the recommended ranges specified in the corresponding references. We detect eight HSIs from two hyperspectral datasets using the proposed method and seven comparative methods. The VNIR and SWIR images of each HSI are subjected to detection, resulting in a total of 16 images for the experiment.

1) GF-5 Dataset: The six VNIR images from the GF-5 dataset contain different types of anomalies with varying backgrounds. As shown in Fig. 11, our proposed method consistently achieves better detection performance on all VNIR images from the GF-5 dataset. Regardless of whether the background is complex or simple and whether the anomalies are large or small, our method effectively suppresses the background while detecting anomalies to the maximum extent. As can be seen, the RX method realizes good detection performance in situations in which the background is relatively simple. However, its detection performance is poor when the background is more complex, such as with GF-5-5 and GF-5-6. For the LRASR, GTVLRR, LSMAD, RGAE, and PTA methods, the anomalies can be highlighted, but the background suppression performance is poor. The spatial–spectral joint-based VABS method can achieve good detection results when the spatial characteristics of the anomalies are prominent. However, when the spatial characteristics of the background in the image are equally prominent, anomalies tend to be missed, resulting in poorer detection performance, such as in GF-5-3.

For the six SWIR images from the GF-5 dataset, the differences between the anomalies and background are smaller than those in the corresponding VNIR images. Some anomalies in certain images, such as GF-5-2 and GF-5-3, are difficult to discern visually. In addition, the striping noise in the SWIR images is more pronounced than that in the VNIR images. These factors impose higher requirements on anomaly detection methods. The detection maps of the different methods on the SWIR images from the GF-5 dataset are shown in Fig. 12. The RX method exhibits a decreased ability to suppress background. Other comparative methods are also significantly affected, losing anomalies in certain images and exhibiting an excessive background response. However, the proposed method remains unaffected and exhibits excellent detection performance. In contrast to the comparative methods, our method achieves the best background suppression while

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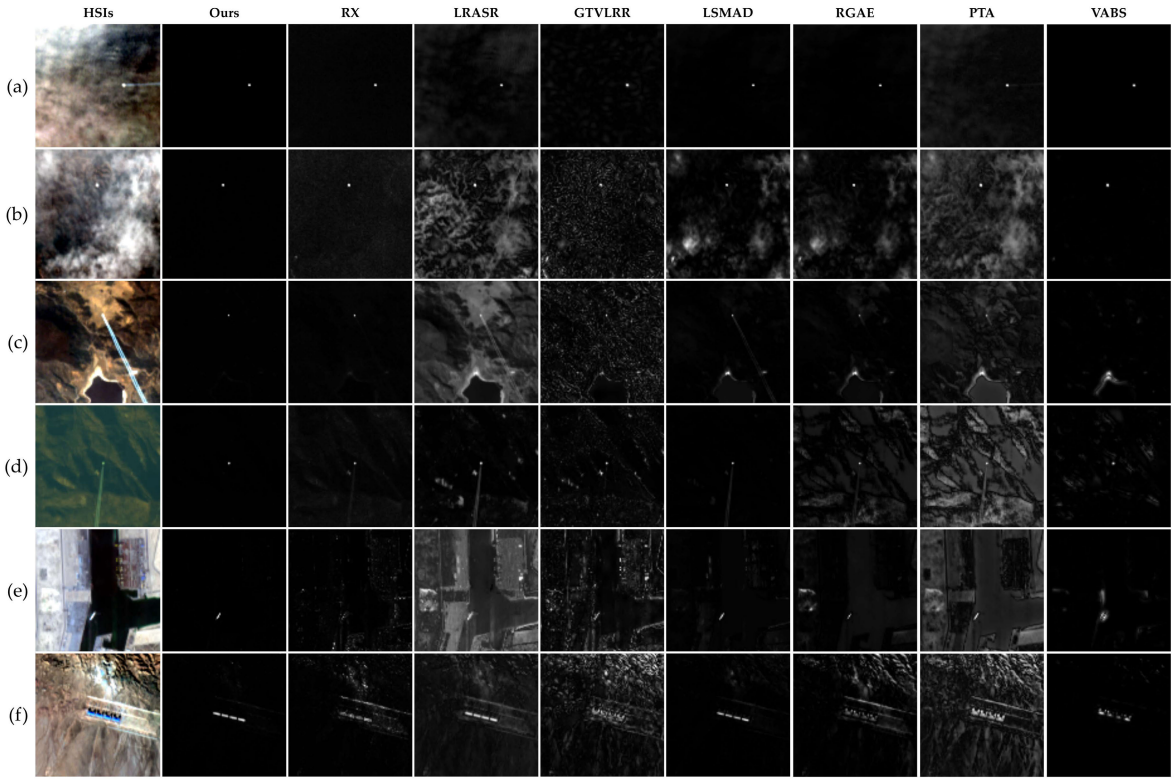


Fig. 11. Detection maps resulting from utilizing different methods on the VNIR images from the GF-5 dataset: (a) GF-5-1, (b) GF-5-2, (c) GF-5-3, (d) GF-5-4, (e) GF-5-5, and (f) GF-5-6.

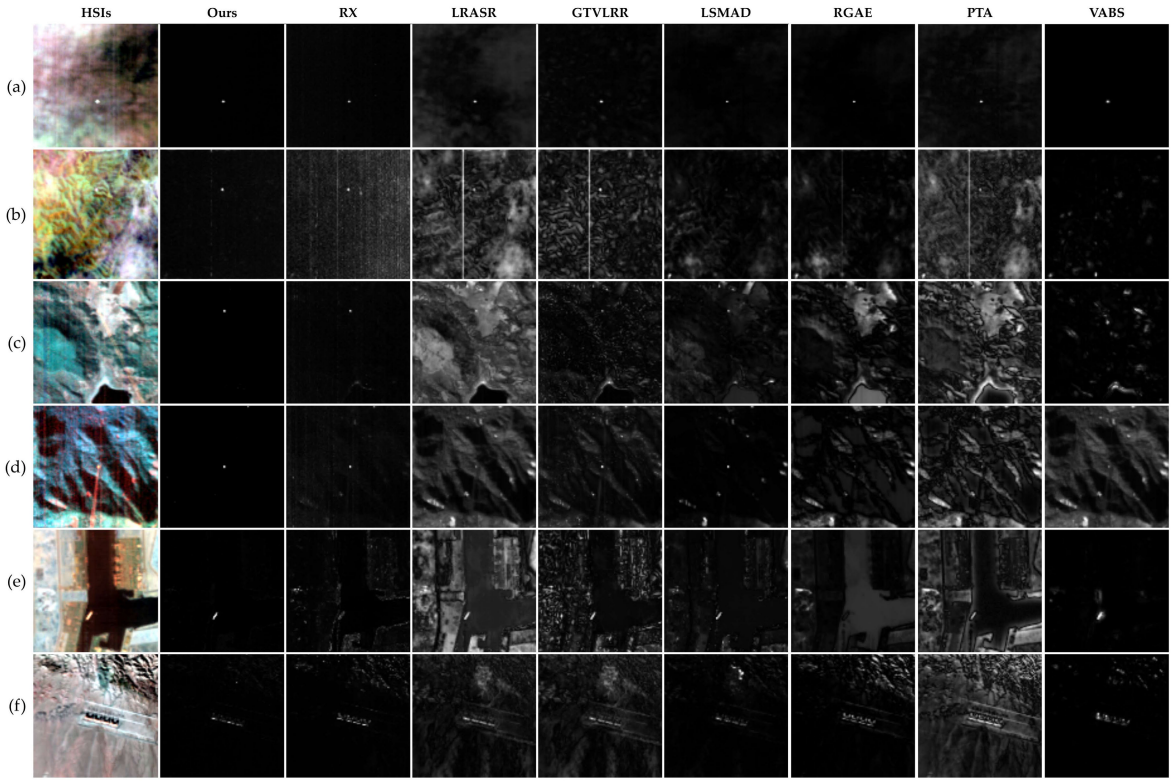


Fig. 12. Detection maps resulting from utilizing different methods on the SWIR images from the GF-5 dataset: (a) GF-5-1, (b) GF-5-2, (c) GF-5-3, (d) GF-5-4, (e) GF-5-5, and (f) GF-5-6.

detecting anomalies in all SWIR images from the GF-5 dataset.

The anomalies in the GF-5-1 and GF-5-2 HSIs are at the pixel level. In the VNIR images, our method and most other

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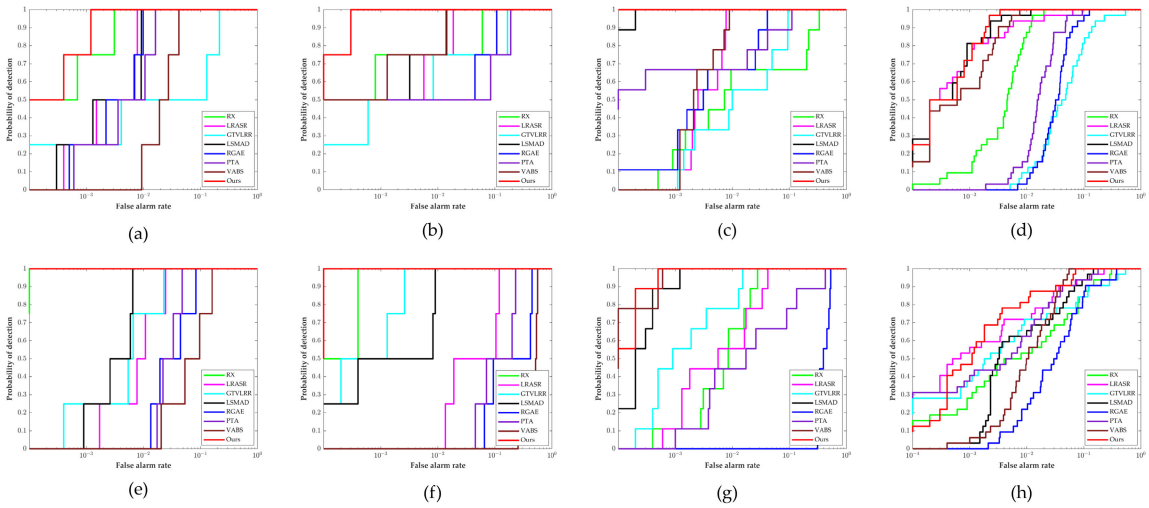


Fig. 13. ROC curves acquired by applying different methods to VNIR and SWIR images from the GF-5 dataset: (a) VNIR image of GF-5-3, (b) VNIR image of GF-5-4, (c) VNIR image of GF-5-5, (d) VNIR image of GF-5-6, (e) SWIR image of GF-5-3, (f) SWIR image of GF-5-4, (g) SWIR image of GF-5-5, and (h) SWIR image of GF-5-6.

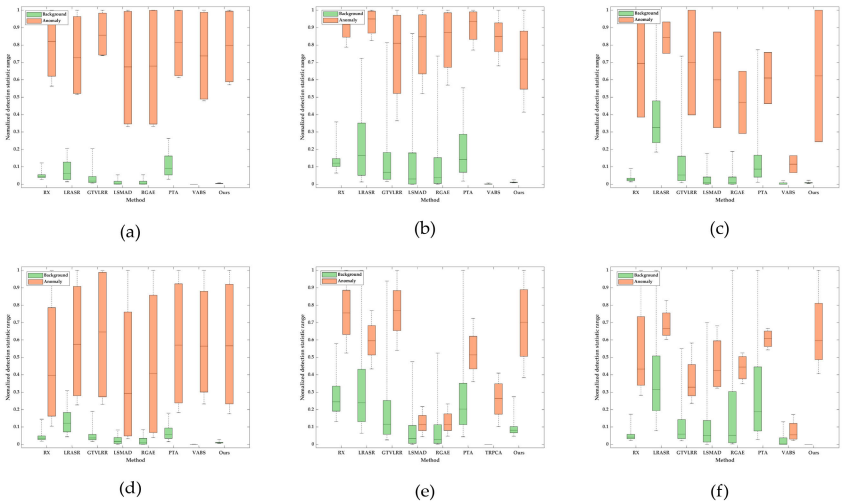


Fig. 14. Anomaly background separability maps obtained using different methods on the VNIR images from the GF-5 dataset: (a) GF-5-1, (b) GF-5-2, (c) GF-5-3, (d) GF-5-4, (e) GF-5-5, and (f) GF-5-6.

comparative methods exhibit the highest responses to anomalous pixels. The corresponding ROC curves form a horizontal line, thus losing their evaluative significance. Therefore, we do not display the ROC curves for these two HSIs. The ROC curves of the different methods for the VNIR and SWIR images of the other four HSIs in the GF-5 dataset are shown in Fig. 13. Our method outperforms the comparative methods in terms of ROC curve performance. It exhibits a higher PD at a lower FAR, with the curve being closer to the top-left corner.

Tables II and III present the AUC values for all VNIR and SWIR images of the HSIs from the GF-5 dataset. The proposed method yields the largest AUC values for all the HSIs. Figs. 14 and 15 show the separability maps for the VNIR and SWIR images of HSIs from the GF-5 dataset. For the VNIR images of GF-5-1 and GF-5-2, although most comparative methods have an AUC value of 1, it can be inferred from the separability maps that their detection performance is not satisfactory because the background boxes in their separability maps cover a wide range, indicating that the background responses are generally distributed at higher levels. However, for the proposed method, the response of the background pixels is nearly zero, and all background boxes approximate a line. The anomaly boxes are located at higher positions, and significant gaps exist between the anomalies and background boxes, indicating that our method has a strong ability to discriminate between anomalies and background. For the comparative methods, the response of the background pixels is generally high and often falls within the range of the anomaly response, indicating a weak ability to distinguish between the background and anomalies. The VABS method exhibits good background suppression capability; however, in certain images, its response to anomalies is too weak to detect.

2) ZY1-02D Dataset: For the ZY1-02D dataset, two HSIs contained two different types of anomalies. In ZY1-02D-1, the anomalies are multiple targets with pixel-level sizes.

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TABLE II
AUC VALUES FOR THE VNIR IMAGES IN THE GF-5 DATASET

Methods	GF-5-1 (VNIR)	GF-5-2 (VNIR)	GF-5-3 (VNIR)	GF-5-4 (VNIR)	GF-5-5 (VNIR)	GF-5-6 (VNIR)
RX	1.0000	1.0000	0.9980	0.9848	0.9137	0.9946
LRASR	1.0000	1.0000	0.9959	0.9939	0.9958	0.9987
GTVLRR	1.0000	0.9989	0.9127	0.9564	0.9665	0.9328
LSMAD	1.0000	0.9982	0.9944	0.9956	0.9998	0.9978
RGAE	1.0000	0.9997	0.9951	0.9620	0.9679	0.9618
PTA	1.0000	1.0000	0.9922	0.9319	0.9882	0.9852
VABS	1.0000	1.0000	0.9755	0.9962	0.9965	0.9478
Ours	1.0000	1.0000	0.9996	0.9999	1.0000	0.9994

TABLE III
AUC VALUES FOR THE SWIR IMAGES IN THE GF-5 DATASET

Method	GF-5-1 (SWIR)	GF-5-2 (SWIR)	GF-5-3 (SWIR)	GF-5-4 (SWIR)	GF-5-5 (SWIR)	GF-5-6 (SWIR)
RX	0.9947	0.9967	0.9999	0.9848	0.9895	0.9552
LRASR	0.9799	0.9615	0.9887	0.9699	0.9867	0.9815
GTVLRR	0.9969	0.9890	0.9912	0.9916	0.9960	0.9469
LSMAD	0.9528	0.9015	0.9960	0.9956	0.9997	0.9771
RGAE	0.9676	0.8676	0.9597	0.9716	0.5876	0.9374
PTA	0.9687	0.9059	0.9698	0.9319	0.9211	0.9801
VABS	0.9992	0.9554	0.9172	0.9265	0.9998	0.9834
Ours	1.0000	0.9999	1.0000	0.9999	1.0000	0.9921

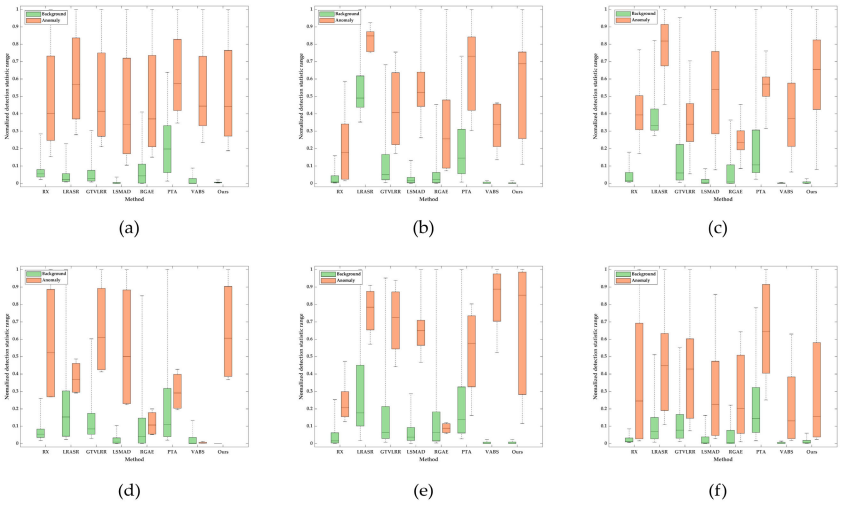


Fig. 15. Anomaly background separability maps acquired by applying different methods to the SWIR images from the GF-5 dataset: (a) GF-5-1, (b) GF-5-2, (c) GF-5-3, (d) GF-5-4, (e) GF-5-5, and (f) GF-5-6.

In ZY1-02D-2, the anomaly is a dark target exhibiting lower radiance than the background. The detection maps of the different methods on the VNIR and SWIR images from the ZY1-02D dataset are shown in Fig. 16. For the ZY1-02D-1, the RX, LSMAD, and VABS methods exhibit low responses to most anomalous pixels, resulting in low detection probabilities. The LRASR, GTVLRR, and RGAE methods show higher responses to anomalies but poor suppression of the background. Our proposed method achieves satisfactory detection results, outperforming the comparative methods in terms of both anomaly detection and background suppression. For the ZY1-02D-2, with a dark anomalous target, the LRASR, GTVLRR, RGAE, and PTA methods are significantly contaminated by the background. Noticeable striping artifacts exist in the detection results of the SWIR images. The VABS method fails to detect anomalous targets in the VNIR image. The RX, LSMAD, and our proposed method produce better detection results; however, the LSMAD method is slightly affected by the stripe noise in the SWIR image.

For the VNIR and SWIR images of ZY1-02D-2, the proposed method and most of the comparative methods exhibit the highest responses to anomalous pixels. The corresponding ROC curves form a horizontal line, thus losing their evaluative significance. Therefore, we do not display the ROC curves

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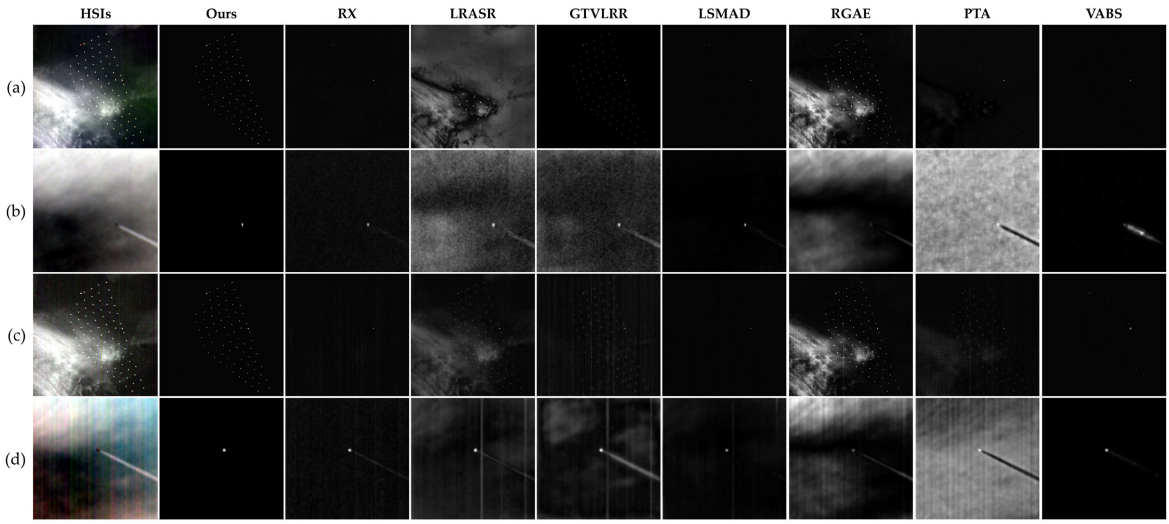


Fig. 16. Detection maps acquired by applying different methods to VNIR and SWIR images from the ZY1-02D dataset: (a) VNIR image of ZY1-02D-1, (b) VNIR image of ZY1-02D-2, (c) SWIR image of ZY1-02D-1, and (d) SWIR image of ZY1-02D-2.

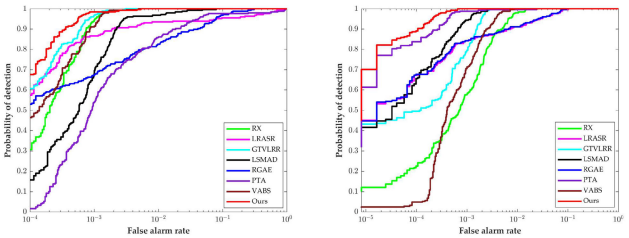


Fig. 17. ROC curves obtained using different methods on ZY1-02D-1: (a) VNIR image of ZY1-02D-1 and (b) SWIR image of ZY1-02D-1.

TABLE IV
AUC VALUES FOR THE VNIR AND SWIR IMAGES IN THE ZY1-02D DATASET

Methods	ZY1-02D-1 (VNIR)	ZY1-02D-2 (VNIR)	ZY1-02D-1 (SWIR)	ZY1-02D-2 (SWIR)
RX	0.9987	1.0000	0.9783	1.0000
LRASR	0.9748	1.0000	0.9910	1.0000
GTVLRR	0.9988	1.0000	0.9886	1.0000
LSMAD	0.9862	1.0000	0.9924	0.9909
RGAE	0.9876	0.4965	0.9912	0.8676
PTA	0.9778	0.9998	0.9951	0.9959
VABS	0.9897	0.9944	0.9769	1.0000
Ours	0.9998	1.0000	0.9999	1.0000

for ZY1-02D-2. Fig. 17 shows the ROC curves obtained by using different methods on ZY1-02D-1. Our proposed method exhibits higher ROC curves than the other methods, achieving detection close to 1 with the lowest false positive rate. Table IV presents the AUC values for the two HSIs from the ZY1-02D dataset; our proposed method exhibits the highest AUC values. The separability maps for the ZY1-02D dataset are presented in Fig. 18. It can be observed that although the RX, LRASR, GTVLRR, and LSMAD methods have AUC values of 1 for the VNIR and SWIR images of ZY1-02D-2, their detection results are significantly contaminated by the background, as inferred from the separability maps in Fig. 18(b) and (d). The large ranges of the background boxes indicate the presence of many background pixels with strong responses. Furthermore, the small gap between the anomaly boxes and background boxes reduces the separability between the anomalies and background. Our method exhibits a significant distance between the anomaly and background boxes in all four images, with the background boxes appearing as lines.

D. Computational Complexity Analysis

Next, we analyze the computational complexity analysis of the proposed Algorithm 2. Let $\boldsymbol{\mathcal{X}} \in \mathbb{R}^{H\times W\times B}$ be the input HSI tensor; then, the computational complexity of each iteration is composed of the following five parts.

1) The computational complexity of updating $\boldsymbol{\mathcal{Z}}^1$ primarily stems from the FFT and SVD. Initially, FFT is performed on the mode-3 fibers of the tensor, where the computational complexity of the FFT for each mode-3 fiber is $O[B\log(B)]$. Consequently, the complexity for applying FFT to all mode-3 fibers of the tensor is $O[HWB\log(B)]$. Subsequently, each frontal slice of the tensor undergoes SVD, with the complexity for decomposing each slice being $O(H^2W)$. Thus, the total complexity for SVD across all frontal slices amounts to $O(H^2WB)$. Therefore, the comprehensive computational complexity of updating $\boldsymbol{\mathcal{Z}}^1$ is $O[H^2WB + HWB\log(B)]$.

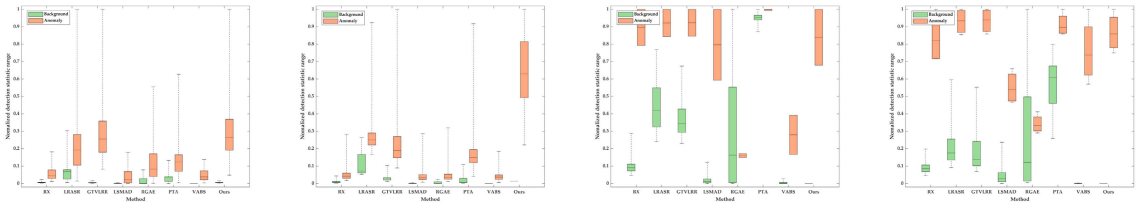
2) Similar to $\boldsymbol{\mathcal{Z}}^1$, the computational complexity of updating $\boldsymbol{\mathcal{Z}}^2$ also stems from the FFT and SVD and is $O[B^2WH + BWH\log(H)]$.

3) The computational complexity of updating $\boldsymbol{\mathcal{Z}}^3$ primarily originates from matrix inversion and matrix multiplication, with a complexity of $O(H^3 + H^2WB)$.

4) Similarly, the computational complexity of updating $\boldsymbol{\mathcal{Z}}^4$ is $O(W^3 + W^2HB)$.

5) For updating variables $\boldsymbol{\mathcal{L}}$ and $\boldsymbol{\mathcal{S}}$ and Lagrange multipliers $\boldsymbol{\mathcal{M}}^1, \boldsymbol{\mathcal{M}}^2, \boldsymbol{\mathcal{M}}^3, \boldsymbol{\mathcal{M}}^4$, and $\boldsymbol{\mathcal{P}}$, the computational complexities are all $O(HWB)$. Thus, the overall computational complexity

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(a) (b) (c) (d)

Fig. 18. Anomaly background separability maps resulting from utilizing different methods on the VNIR and SWIR images from the ZY1-02D dataset: (a) VNIR image of ZY1-02D-1, (b) VNIR image of ZY1-02D-2, (c) SWIR image of ZY1-02D-1, and (d) SWIR image of ZY1-02D-2.

TABLE V
AVERAGE EXECUTION TIME (IN SECONDS) OF DIFFERENT METHODS ON THE GF-5 AND ZY1-02D DATASETS

Methods	RX	LRASR	GTVLRR	LSMAD	RGAE	PTA	VABS	Ours
GF-5 VNIR	0.0476	7.7970	132.5318	4.6128	141.7889	23.4626	4.0845	44.9769
GF-5 SWIR	0.0596	9.9328	160.7251	6.9077	180.8237	33.5188	7.8465	62.4085
ZY-1-02D VNIR	0.0189	7.7321	112.2274	2.7634	105.3986	14.2638	4.0318	27.7525
ZY-1-02D SWIR	0.0224	7.8956	118.7644	3.1753	113.8922	17.6346	7.2749	32.4336

of our solving algorithm is $O(H^3 + W^3 + HWB + H^2WB + W^2HB + B^2WH + HWB[\log(B) + \log(H)])$.

To further quantitatively analyze algorithm computational complexity, we present the average execution time of our algorithm and the comparative algorithms on two HSI datasets, as shown in Table V. For iterative methods such as LRASR, GTVLRR, LSMAD, PTA, and the proposed method, we have set the number of iterations to 100. It can be observed that our method does not hold an advantage in terms of time consumption, being slower than RX, LRASR, LSMAD, VABS, and PTA methods. However, it is still faster than the GTVLRR and RGAE methods. All experiments are conducted on a computer with an AMD Ryzen 7 4800H CPU at 2.90 GHz and 16 GB of memory.

In summary, our method fully incorporates the inherent physical characteristics of HSIs obtained through satellite-borne push-broom imaging technology. We design a joint low-rank regularization term to characterize the low-rank properties in both spatial and spectral dimensions simultaneously and flexibly, resulting in a more accurate approximation of the background tensor, thus achieving improved anomaly detection in real HSIs. Specifically, our method can detect anomalous targets with high accuracy while effectively suppressing the background and demonstrating low sensitivity to noise. It outperforms several compared common anomaly detection methods across multiple evaluation metrics, such as detection maps, ROC curves, AUC values, and separability maps. These are the advantages of the method that we propose; however, there are still some shortcomings that need to be addressed in future research. As discussed above, our proposed method has a relatively high computational complexity, which limits its real-time application. In addition, our method relies on the manual setting of three parameters. Improving it to an end-to-end anomaly detection method with adaptive parameter adjustment is an important direction for future research.

VI. CONCLUSION

In this study, we propose an anomaly detection method based on JLRTA for HSIs obtained through satellite-borne push-broom imaging. First, we assumed that the original hyperspectral tensor could be decomposed into the sum of the background and anomaly tensors. Considering the inherent spatial–spectral low-rank characteristics of HSIs acquired through satellite-borne push-broom imaging, a joint low-rank regularization term based on weighted tensor nuclear norm and tensor permutation operation is designed to impose low-rank constraints on background tensor in both spatial and spectral dimensions simultaneously and flexibly. Furthermore, to minimize the chances of missing anomalies, a linear TV-norm regularization term is incorporated into the model to constrain the spatial piecewise smoothness of the background tensor. This enables the background tensor to contain fewer spatially discontinuous anomalies. For the anomaly tensor, an $\ell_{1,1,2}$-norm regularization term is employed to constrain the tubewise sparsity. All regularization terms and a fidelity term are incorporated into a nonconvex framework, and the solving algorithm is achieved using the ADMM. Finally, the optimal anomaly tensor is obtained and utilized to generate the final anomaly map. Extensive experimental results demonstrate that the proposed method outperforms seven other comparative methods on real HSIs from the GF-5 and ZiYuan1-02D satellites.

REFERENCES

[1] N. Audebert, B. Le Saux, and S. Lefevre, "Deep learning for classification of hyperspectral data: A comparative review," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 159–173, Jun. 2019.

[2] H. Zhang, Y. Li, Y. Jiang, P. Wang, Q. Shen, and C. Shen, "Hyperspectral classification based on lightweight 3-D-CNN with transfer learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5813–5828, Aug. 2019.

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[3] F. A. Kruse, J. W. Boardman, and J. F. Huntington, "Comparison of airborne hyperspectral data and EO-1 hyperion for mineral mapping," *IEEE Trans. Geosci. Remote Sens.*, vol. 41, no. 6, pp. 1388–1400, Jun. 2003.

[4] C. A. Bishop, J. G. Liu, and P. J. Mason, "Hyperspectral remote sensing for mineral exploration in Pulang, Yunnan Province, China," *Int. J. Remote Sens.*, vol. 32, no. 9, pp. 2409–2426, May 2011.

[5] I. Irakulis-Loitxate et al., "Satellite-based survey of extreme methane emissions in the Permian basin," *Sci. Adv.*, vol. 7, no. 27, Jul. 2021, Art. no. eabf4507.

[6] E. D. Sherwin, Y. Chen, A. P. Ravikumar, and A. R. Brandt, "Single-blind test of airplane-based hyperspectral methane detection via controlled releases," *Elementa, Sci. Anthropocene*, vol. 9, no. 1, Mar. 2021, Art. no. 00063.

[7] Y. Zhang, B. Du, L. Zhang, and T. Liu, "Joint sparse representation and multitask learning for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 2, pp. 894–906, Feb. 2017.

[8] Y. Dong, W. Shi, B. Du, X. Hu, and L. Zhang, "Asymmetric weighted logistic metric learning for hyperspectral target detection," *IEEE Trans. Cybern.*, vol. 52, no. 10, pp. 11093–11106, Oct. 2022.

[9] N. Huyan, X. Zhang, H. Zhou, and L. Jiao, "Hyperspectral anomaly detection via background and potential anomaly dictionaries construction," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2263–2276, Apr. 2019.

[10] L. Li, W. Li, Q. Du, and R. Tao, "Low-rank and sparse decomposition with mixture of Gaussian for hyperspectral anomaly detection," *IEEE Trans. Cybern.*, vol. 51, no. 9, pp. 4363–4372, Sep. 2021.

[11] I. S. Reed and X. Yu, "Adaptive multiple-band CFAR detection of an optical pattern with unknown spectral distribution," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. 38, no. 10, pp. 1760–1770, Oct. 1990, doi: 10.1109/29.60107.

[12] J. M. Molero, E. M. Garzón, I. García, and A. Plaza, "Analysis and optimizations of global and local versions of the RX algorithm for anomaly detection in hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 6, no. 2, pp. 801–814, Apr. 2013, doi: 10.1109/JSTARS.2013.2238609.

[13] C.-I. Chang and S.-S. Chiang, "Anomaly detection and classification for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 40, no. 6, pp. 1314–1325, Jun. 2002, doi: 10.1109/TGRS.2002.800280.

[14] H. Kwon and N. M. Nasrabadi, "Kernel RX-algorithm: A nonlinear anomaly detector for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 2, pp. 388–397, Feb. 2005, doi: 10.1109/TGRS.2004.841487.

[15] B. Du and L. Zhang, "Random-selection-based anomaly detector for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 5, pp. 1578–1589, May 2011, doi: 10.1109/TGRS.2010.2081677.

[16] Q. Guo, B. Zhang, Q. Ran, L. Gao, J. Li, and A. Plaza, "Weighted-RXD and linear filter-based RXD: Improving background statistics estimation for anomaly detection in hyperspectral imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2351–2366, Jun. 2014, doi: 10.1109/JSTARS.2014.2302446.

[17] S. Lin, M. Zhang, X. Cheng, K. Zhou, S. Zhao, and H. Wang, "Hyperspectral anomaly detection via sparse representation and collaborative representation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 946–961, 2023.

[18] J. Li, H. Zhang, L. Zhang, and L. Ma, "Hyperspectral anomaly detection by the use of background joint sparse representation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2523–2533, Jun. 2015.

[19] Q. Ling, Y. Guo, Z. Lin, and W. An, "A constrained sparse representation model for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2358–2371, Apr. 2019, doi: 10.1109/TGRS.2018.2872900.

[20] W. Li and Q. Du, "Collaborative representation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 3, pp. 1463–1474, Mar. 2015, doi: 10.1109/TGRS.2014.2343955.

[21] W. Li, G. Wu, and Q. Du, "Transferred deep learning for anomaly detection in hyperspectral imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 5, pp. 597–601, May 2017, doi: 10.1109/LGRS.2017.2657818.

[22] S. Wang, X. Wang, L. Zhang, and Y. Zhong, "Auto-AD: Autonomous hyperspectral anomaly detection network based on fully convolutional autoencoder," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 3057721, doi: 10.1109/TGRS.2021.3057721.

[23] G. Fan, Y. Ma, X. Mei, F. Fan, J. Huang, and J. Ma, "Hyperspectral anomaly detection with robust graph autoencoders," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 3097097, doi: 10.1109/TGRS.2021.3097097.

[24] T. Jiang, Y. Li, W. Xie, and Q. Du, "Discriminative reconstruction constrained generative adversarial network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4666–4679, Jul. 2020, doi: 10.1109/TGRS.2020.2965961.

[25] T. Jiang, W. Xie, Y. Li, J. Lei, and Q. Du, "Weakly supervised discriminative learning with spectral constrained generative adversarial network for hyperspectral anomaly detection," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 11, pp. 6504–6517, Nov. 2022.

[26] W. Sun, C. Liu, J. Li, Y. M. Lai, and W. Li, "Low-rank and sparse matrix decomposition-based anomaly detection for hyperspectral imagery," *J. Appl. Remote Sens.*, vol. 8, no. 1, May 2014, Art. no. 083641, doi: 10.1117/1.Jrs.8.083641.

[27] Y. Zhang, B. Du, L. Zhang, and S. Wang, "A low-rank and sparse matrix decomposition-based Mahalanobis distance method for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 3, pp. 1376–1389, Mar. 2016, doi: 10.1109/TGRS.2015.2479299.

[28] Y. Xu, Z. Wu, J. Li, A. Plaza, and Z. Wei, "Anomaly detection in hyperspectral images based on low-rank and sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 4, pp. 1990–2000, Apr. 2016, doi: 10.1109/TGRS.2015.2493201.

[29] T. Cheng and B. Wang, "Graph and total variation regularized low-rank representation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 391–406, Jan. 2020, doi: 10.1109/TGRS.2019.2936609.

[30] L. Li, W. Li, Y. Qu, C. Zhao, R. Tao, and Q. Du, "Prior-based tensor approximation for anomaly detection in hyperspectral imagery," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 3, pp. 1037–1050, Mar. 2022.

[31] T. G. Kolda and B. W. Bader, "Tensor decompositions and applications," *SIAM Rev.*, vol. 51, no. 3, pp. 455–500, Aug. 2009.

[32] L. Tucker, "Implications of factor analysis of three-way matrices for measurement of change," in *Problems in Measuring Change*. Madison, WI, USA: Univ. Wisconsin Press, 1963.

[33] R. A. Harshman, "Foundations of the PARAFAC procedure: Models and conditions for an 'explanatory' multimodal factor analysis," in *Proc. UCLA Working Papers Phonetics*, vol. 16, 1970, pp. 1–84.

[34] M. E. Kilmer and C. D. Martin, "Factorization strategies for third-order tensors," *Linear Algebra Appl.*, vol. 435, no. 3, pp. 641–658, Aug. 2011.

[35] M. Zhao, W. Li, L. Li, P. Ma, Z. Cai, and R. Tao, "Three-order tensor creation and tucker decomposition for infrared small-target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 3057696.

[36] J. Xue, Y. Zhao, S. Huang, W. Liao, J. C. Chan, and S. G. Kong, "Multilayer sparsity-based tensor decomposition for low-rank tensor completion," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 11, pp. 6916–6930, Nov. 2022.

[37] S. Li, W. Wang, H. Qi, B. Ayhan, C. Kwan, and S. Vance, "Low-rank tensor decomposition based anomaly detection for hyperspectral imagery," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2015, pp. 4525–4529.

[38] W. Shang, M. Jouni, Z. Wu, Y. Xu, M. D. Mura, and Z. Wei, "Hyperspectral anomaly detection based on regularized background abundance tensor decomposition," *Remote Sens.*, vol. 15, no. 6, p. 1679, Mar. 2023.

[39] M. E. Kilmer, K. Braman, N. Hao, and R. C. Hoover, "Third-order tensors as operators on matrices: A theoretical and computational framework with applications in imaging," *SIAM J. Matrix Anal. Appl.*, vol. 34, no. 1, pp. 148–172, Jan. 2013.

[40] R. A, X. Mu, and J. He, "Enhance tensor RPCA-based Mahalanobis distance method for hyperspectral anomaly detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[41] Y. Xu, Z. Wu, J. Chanussot, and Z. Wei, "Joint reconstruction and anomaly detection from compressive hyperspectral images using Mahalanobis distance-regularized tensor RPCA," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 5, pp. 2919–2930, May 2018.

[42] Z. Zhang, G. Ely, S. Aeron, N. Hao, and M. Kilmer, "Novel methods for multilinear data completion and de-noising based on tensor-SVD," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 3842–3849.

[43] O. Semerci, N. Hao, M. E. Kilmer, and E. L. Miller, "Tensor-based formulation and nuclear norm regularization for multienergy computed tomography," *IEEE Trans. Image Process.*, vol. 23, no. 4, pp. 1678–1693, Apr. 2014.

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[44] Y. Liu et al., "Development of visible and short-wave infrared hyperspectral imager onboard GF-5 satellite," *Nat. Remote Sens. Bull.*, vol. 24, no. 4, pp. 333–344, 2020.

[45] Y.-N. Liu et al., "The advanced hyperspectral imager: Aboard China's GaoFen-5 satellite," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 4, pp. 23–32, Dec. 2019.

[46] Y. Liu, D. Sun, B. Han, H. Zhu, S. Liu, and J. Yuan, "Development of advanced visible and short-wave infrared hyperspectral imager onboard ZY-1-02D satellite," *Spacecr. Eng.*, vol. 29, no. 6, pp. 85–92, 2020, doi: 10.3969/j.issn.1673-8748.2020.06.012.

[47] N. Renard, S. Bourennane, and J. Blanc-Talon, "Denoising and dimensionality reduction using multilinear tools for hyperspectral images," *IEEE Geosci. Remote Sens. Lett.*, vol. 5, no. 2, pp. 138–142, Apr. 2008.

[48] Y.-B. Zheng, T.-Z. Huang, X.-L. Zhao, T.-X. Jiang, T.-H. Ma, and T.-Y. Ji, "Mixed noise removal in hyperspectral image via low-fibered-rank regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 734–749, Jan. 2020, doi: 10.1109/TGRS.2019.2940534.

[49] S. Gu, L. Zhang, W. Zuo, and X. Feng, "Weighted nuclear norm minimization with application to image denoising," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 2862–2869.

[50] L. Zhai, S. Fu, H. Lv, C. Zhang, and F. Wang, "Weighted schatten p-norm minimization for 3D magnetic resonance images denoising," *Brain Res. Bull.*, vol. 142, pp. 270–280, Sep. 2018.

[51] Y. Mu, P. Wang, L. Lu, X. Zhang, and L. Qi, "Weighted tensor nuclear norm minimization for tensor completion using tensor-SVD," *Pattern Recognit. Lett.*, vol. 130, pp. 4–11, Feb. 2020.

[52] M.-D. Iordache, J. M. Bioucas-Dias, and A. Plaza, "Total variation spatial regularization for sparse hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 11, pp. 4484–4502, Nov. 2012.

[53] L. I. Rudin, S. Osher, and E. Fatemi, "Nonlinear total variation based noise removal algorithms," *Phys. D, Nonlinear Phenomena*, vol. 60, nos. 1–4, pp. 259–268, Nov. 1992.

[54] F. Nie, H. Huang, X. Cai, and C. Ding, "Efficient and robust feature selection via joint $\ell_{2,1}$-norms minimization," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 23, 2010, pp. 1813–1821.

[55] S. Boyd, "Distributed optimization and statistical learning via the alternating direction method of multipliers," *Found. Trends Mach. Learn.*, vol. 3, no. 1, pp. 1–122, 2010.

[56] J.-F. Cai, E. J. Candès, and Z. Shen, "A singular value thresholding algorithm for matrix completion," *SIAM J. Optim.*, vol. 20, no. 4, pp. 1956–1982, Jan. 2010.

[57] Q. Gao, W. Xia, Z. Wan, D. Xie, and P. Zhang, "Tensor-SVD based graph learning for multi-view subspace clustering," in *Proc. AAAI Conf. Artificial Intell.*, 2020, pp. 3930–3937.

[58] S. Sun, J. Liu, Z. Zhang, and W. Li, "Hyperspectral anomaly detection based on adaptive low-rank transformed tensor," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, doi: 10.1109/TNNLS.2023.3236641.

[59] J. Kerekes, "Receiver operating characteristic curve confidence intervals and regions," *IEEE Geosci. Remote Sens. Lett.*, vol. 5, no. 2, pp. 251–255, Apr. 2008.

[60] P. Xiang, J. Song, H. Qin, W. Tan, H. Li, and H. Zhou, "Visual attention and background subtraction with adaptive weight for hyperspectral anomaly detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 2270–2283, 2021, doi: 10.1109/JSTARS.2021.3052968.



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